

ESG FEDERALISM: HOW STATE
REGULATIONS SHAPE ESG PERFORMANCE

A Thesis

Presented to

The Graduate Faculty

Central Washington University

In Partial Fulfillment

of the Requirements for the Degree

Master of Science

Cultural and Environmental Resource Management

by

Kameron Eck

March 2025

CENTRAL WASHINGTON UNIVERSITY

Graduate Studies

We hereby approve the thesis of

Kameron Eck

Candidate for the degree of Master of Science

APPROVED FOR THE GRADUATE FACULTY

Toni Sipic, Committee Chair

Kevin Archer

Han Donker

Dean of Graduate Studies

ABSTRACT

ESG FEDERALISM: HOW STATE REGULATIONS SHAPE ESG PERFORMANCE

By

Kameron Eck

March 2024

This thesis examines how state-level environmental, social, and governance (ESG) regulations influence corporate ESG performance in the United States. Using a panel dataset that combines state ESG legislation, firm-level ESG scores from S&P Global, and financial data from Compustat (2019–2024), I employ fixed-effects regression models to analyze the relationship between state ESG regulations and corporate ESG performance. The analysis incorporates firm and year fixed effects to account for unobservable heterogeneity across companies and time, while controlling for firm-specific characteristics. Additionally, the study investigates whether board gender diversity moderates this relationship. The findings contribute to the understanding of ESG policy effectiveness and whether such regulations serve as substantive mechanisms for impacting corporate sustainability.

ACKNOWLEDGEMENTS

I extend my deepest gratitude to Dr. Toni Sipic for nurturing my curiosity and constantly challenging me to push forward, even when the path was uncertain. Your unwavering encouragement has been invaluable in shaping both my intellectual and personal growth. I am also profoundly grateful to Dr. Kevin Archer, whose wisdom provided the foundation that allowed me to connect ideas and perspectives in meaningful ways. Additionally, I would like to express my sincere appreciation to Dr. Han Donker for their guidance and support throughout this journey.

TABLE OF CONTENTS

CHAPTER		PAGE
I	INTRODUCTION	1
	The Rise of Corporate Social Responsibility and ESG	1
	The Institutionalization of ESG	2
	The Emerging Countermovement.....	3
	Research Focus	4
II	LITERATURE REVIEW	5
	ESG Data	5
	Theory and ESG.....	7
	European Union ESG Legislation.....	9
	United States ESG Legislation.....	10
	Firm Determinants of ESG Performance.....	17
	Regulatory Frameworks as Determinants of ESG Performance.....	19
	Industry Classification	22
III	DATA	24
	ESG Data	24
	Board Data	25
	Firm Data	26
	Legislative Data	27
	Sample Construction.....	29
IV	JOURNAL ARTICLE	31
	Introduction.....	31
	Literature Review.....	32
	Data	41
	Empirical Method	49
	Discussion	51
V	CONCLUSION.....	58
VI	REFERENCES	61
VII	APPENDICIES	78
	Appendix A.....	78
	Appendix B	86

LIST OF TABLES

TABLE		PAGE
1	S&P Global ESG Indices Calculations	25
2	S&P Global Gender Diversity and Tenure Indices Calculations.....	25
3	Control Variable Calculations.....	26
4	Summary Statistics.....	30
5	Baseline Results	53
6	Interaction Results	56

LIST OF FIGURES

FIGURE		PAGE
1	State ESG Legislation Trends (2020-2024)	27
2	State-Level ESG Regulatory Regimes (2020-2024)	29
3	ESG Legislation Data Pipeline.....	46

CHAPTER I

INTRODUCTION

The foundations of modern corporate philosophy trace back to Adam Smith's landmark work *The Wealth of Nations* (1776), which established that open and free competition drives economic prosperity. Smith argued that global trade, enhanced labor productivity, and efficient resource allocation collectively function to lower costs and benefit society. This classical view positioned the pursuit of profit within a framework of natural market dynamics that would theoretically benefit all participants through the "invisible hand" of the market.

Nearly two centuries later, Milton Friedman further crystallized this approach in his influential 1970 *New York Times* article, where he declared that "there is one and only one social responsibility of business—to use its resources and engage in activities designed to increase its profits so long as it stays within the rules of the game, which is to say, engages in open and free competition without deception or fraud." (Friedman, 1957). This profit maximization doctrine became the dominant paradigm for corporate behavior throughout much of the late 20th century, providing clear guidance for executives and shareholders alike.

The Rise of Corporate Social Responsibility and ESG

The 1987 Brundtland Report introduced the concept of sustainability as "meeting the needs of the present without compromising the ability of future generations to meet their own needs" (World Commission on Environment and Development, 1987). Sustainability is an economic development theory that directly conflicts with maximization theory. This is exemplified by the 1990's Nike Sweatshop Scandal, where Pakistani children earned just 6 cents an hour sewing soccer balls, and malnourished women worked 65-hour weeks for only \$1 an hour (Ballinger, 1992; Schanberg, 1996).

These practices directly conflict with consumer and investor ethical standards, and Nike's bottom line (Bartley & Child, 2011). To rebuild its reputation, Nike established a corporate social responsibility department and initiated workplace and health audits at its factories, enlisting the Fair Labor Association to conduct random facility inspections (Fair Labor Association, 2003; Zadek, 2007). The essence of Nike's behavior is explained by signaling theory, whereby firms publish information to gain affinity from stakeholders (Friske et al., 2023). By disclosing information about its environmental and social misconduct, establishing accountability measures, and creating avenues for change and employee reparations, Nike was able to rebuild much of its lost customer and investor confidence (Murphy & Mathew, 2001; Achabou, 2020).

The Institutionalization of ESG

The Nike scandal put the international business community on notice. Recognizing the opportunity, the United Nations leader Kofi Annan called upon the global finance community to join the United Nations, whereby ethical values would be injected into the faceless, cold, calculated global economy (Williams, 2007). In 2004, a UN titled report "Who Cares Wins" released recommendations to implement integrate environmental, social and governance (ESG) factors into analysis, asset management and securities brokerage (UN Environment Programme – Finance Initiative, 2004).

The 2004 ESG framework blossomed into a multi-continental endeavor whereby private firms known as data providers actively observe and compile the ESG footprints of corporations. In turn, these data providers sell their data to buyers. The data has been hotly contested, with critics pointing out that ESG is subjective and difficult to measure (Eccles et al., 2020). No complete solution for subjectivity exists at the time of this study, but efforts to standardize ESG

and improve its validity are ongoing. The EU's latest effort to improve data quality came in the form of a 2024 directive that mandates methodology disclosure to regulators (European Parliament, 2022; Annunziata & Siri, 2025). ESG regulation is not strictly limited to the EU. In 2021, China's Securities Regulatory Commission (CSRC) adopted an ESG disclosure mandate for public firm (Ministry of Ecology and Environment, 2021; Shen et al., 2023).

Normative arguments advocating for ESG integration have historically been insufficient, but government ESG regulations signal a changing systemic condition, whereby ESG factors are becoming part of the profit maximization cost basis (Alexander, 2007). This allows profit seeking to operate within the ESG boundaries that sustainable development encourages (Friede et al., 2015; D. Z. X. Huang, 2021).

The Emerging Countermovement

While the EU and China are gravitating towards sustainable development and ESG, the American Federalist system is exhibiting a divergent ESG condition. Along political party lines, states are introducing and passing a significant amount of Pro and Anti-ESG legislation. For instance, Texas Senate Bill 19 represent a significant countermovement against ESG investing, effectively barring financial institutions with ESG commitments from participating in government contract bidding (Garrett & Ivanov, 2022; Texas SB 13, 2021;). This legislative action has produced measurable economic consequences for the state. The exclusion of competitive institutional bidders like Citibank has decreased market competition, ultimately increasing borrowing costs for Texas taxpayers by approximately \$400 million annually (Garrett & Ivanov, 2022).

Research Focus

While Texas provides a compelling case study of the immediate financial impact of anti-ESG policies, this research aims to broaden the analysis by examining how both pro-ESG and anti-ESG legislation across multiple states affects corporate ESG performance. By doing so, this study will complement Garrett and Ivanov's economic analysis with crucial insights into the social and ecological implications of these policy positions, providing a more comprehensive understanding of how state-level ESG policies shape corporate behavior and sustainability outcomes.

The remainder of this thesis is structured as follows. Chapter II presents a comprehensive literature review that situates this research within the existing academic discourse. Chapter III describes the data. Chapter VI consists of a complete journal article, which includes its own introduction, literature review, data description, empirical methodology, discussion of findings, and conclusion. Chapter V outlines potential directions for future research, highlighting unexplored areas and questions that emerged from this study but remain beyond its scope.

CHAPTER II

LITERATURE REVIEW

This chapter reviews the literature on (1) ESG data, (2) ESG theory, (3) European Union ESG legislation, (4) United States ESG legislation, (5) firm determinants of ESG performance, (6) regulatory frameworks as determinants of ESG performance, and (7) industry classification. It begins by examining how ESG data is collected, reported, and standardized. The discussion then addresses foundational ESG theory, followed by overviews of EU and U.S. ESG legislation. Next, it explores the firm-level drivers of ESG performance and the broader impact of regulatory contexts. The section concludes with an analysis of how industry classification influences sustainability outcomes.

ESG Data

The origins and development of ESG as a data product can be traced through a combination of historical context, evolving methodologies, and the increasing demand for non-financial metrics in investment decision-making. ESG has emerged as a critical framework for assessing corporate behavior and sustainability, significantly influencing investment strategies and corporate governance.

Initially, the concept of ESG began to gain traction in the early 2000s, driven by a growing awareness of corporate social responsibility (CSR) and the need for sustainable business practices. This shift was partly a response to various global challenges, including climate change, social inequality, and governance scandals, which highlighted the limitations of traditional financial metrics in capturing a company's long-term viability and societal impact (Amel-Zadeh & Serafeim, 2018). As companies began to integrate ESG criteria into their operations, the demand for standardized ESG data grew, leading to the establishment of various ESG rating

agencies and frameworks that sought to quantify and assess ESG performance (Avetisyan & Hockerts, 2017; Escrig-Olmedo et al., 2019).

The evolution of ESG as a data product has been marked by the development of methodologies for collecting and analyzing ESG data. Initially, ESG assessments relied heavily on self-reported data from companies, which often led to issues of reliability and transparency (Jonsdottir et al., 2022). However, as the ESG landscape matured, the integration of third-party data sources, including news reports, social media sentiment, and regulatory filings, began to enhance the robustness of ESG metrics (De Giuli et al., 2024). This transition has been facilitated by advancements in technology, particularly the adoption of Industry 4.0 technologies such as big data analytics and artificial intelligence, which have improved the accuracy and efficiency of ESG data collection and analysis (Saxena et al., 2022).

Moreover, the rise of ESG-focused investment products has further propelled the development of ESG data. Institutional investors increasingly recognize the financial materiality of ESG factors, leading to a surge in demand for ESG-compliant investment vehicles (Amel-Zadeh & Serafeim, 2018). This trend has prompted financial institutions to refine their ESG assessment methodologies, ensuring that they align with evolving regulatory frameworks and investor expectations (L. Wang, 2024).

However, significant challenges persist, including the need for greater consistency in ESG reporting standards and the reduction of firms' tendencies to exaggerate their ESG commitments to enhance market perceptions (Chen et al., 2023). The regulatory environment that enabled greenwashing has also contributed to discrepancies in ESG evaluations, where two separate firms assessing the same company's ESG performance may arrive at divergent ESG scores (Berg et al., 2022). Due to the complex and context-specific nature of ESG factors, rating

agencies often employ distinct interpretations of ESG the framework influenced by their social origins and institutional contexts, leading to inconsistent scores (Chatterji et al., 2016). These framework implementations reflect the agencies' differing interpretations of materiality and contrasting perspectives on the purpose of ESG data.

Theory and ESG

Signaling Theory

Signaling theory plays a crucial role in understanding how firms communicate critical information to stakeholders. Rooted in the work of Spence (1973, 2002), this theory posits that organizations signal their quality, reliability, and intentions through observable actions and strategic behaviors. At its core, signaling theory addresses the issue of information asymmetry, where firms (as information holders) possess knowledge that stakeholders (as information receivers) lack (Connelly et al., 2011). By effectively leveraging signals, firms can influence stakeholder perceptions, reduce uncertainty, and enhance trust in their market positioning. Thus, any transactions that occur will be more informed and ideally beneficial to the firm providing the signals (Spence, 2002).

In the context of ESG, signaling theory has been increasingly applied to understand how and why firms use ESG disclosures to communicate with stakeholders (Mervelskemper & Streit, 2017; Xie et al., 2019). Mervelskemper and Streit (2017) find that capital market investors assign greater value to ESG performance when firms integrate financial and ESG disclosures into a single report, resulting in superior market valuation effects. Similarly, Xie et al. (2019) demonstrate that firms providing a moderate level of ESG disclosure do not suffer adverse market performance. A key tenet of signaling theory is its role in conveying unobservable but materially significant firm attributes to stakeholders (Connelly et al., 2011). However, routine

disclosures, such as annual ESG reports, may diminish signaling effectiveness if they fail to introduce new, unobservable information (Connelly et al., 2011; Xie et al., 2019). Expanding on this notion, research suggests that firms with no prior history of ESG disclosures experience significantly higher market valuations upon their first ESG report, highlighting the importance of both signal novelty and quality (K. T. Wang & Li, 2016).

Critical Mass Theory

Critical Mass Theory, originally developed by Granovetter (1978) and Olson (1965), asserts that a minimum representation from a group is necessary for these individuals to influence group dynamics and decision-making effectively. Granovetter's (1978) threshold model of collective behavior highlights how individuals' participation in collective action depends on the number of others already engaged, which aligns with the notion that minorities within a group may require a critical mass to drive meaningful change. Kanter (1977) expanded this concept in the corporate context, arguing that when women are significantly underrepresented on boards, they are often seen as tokens with limited influence. She suggested that when a critical mass typically three or more women is reached can they shift boardroom dynamics and be perceived as individuals rather than representatives of their gender.

Konrad et al. (2008) empirically supported the concept of critical mass, demonstrating that boards with three or more women exhibited improved governance, decision-making, and overall effectiveness. Their findings suggest that reaching this threshold enables women to contribute more freely, mitigating tokenism and fostering a more inclusive boardroom dynamic. Building on this, Yarram and Adapa (2021) provided further evidence for both token theory and critical mass theory in their study of ASX 300 companies. Their research indicates that minimal female representation on boards has no significant impact on corporate social responsibility

(CSR), likely due to the challenges of being a lone voice in a male-dominated setting. However, when companies achieve a more balanced gender composition, CSR performance significantly improves. Yarram and Adapa tested thresholds of one, two, and three female directors, finding that boards with three or more women were positively and significantly associated with Total CSR, with an effect size of 5.771 points and a probability value of 0.05.

European Union ESG Legislation

The European Union's efforts to enhance ESG disclosures have progressed through a series of directives aimed at improving transparency and accountability. The first major step was the 2014 Non-Financial Reporting Directive (NFRD), which required large public-interest entities to disclose information on environmental issues, social matters, human rights, anti-corruption practices, and diversity policies (Arif et al., 2022; Lombardi et al., 2022; Matuszak & Różańska, 2021). Empirical research shows that this mandate substantially increased the volume of non-financial disclosures, particularly among firms with little to no prior ESG reporting (Aluchna et al., 2023; Čufar & Belak, 2024; Cuomo et al., 2024). However, the lack of standardized frameworks and the variability of national regulations led to inconsistent compliance across EU member states (Cicchiello et al., 2023; Hao et al., 2023).

To address these inconsistencies, the European Commission introduced the Corporate Sustainability Reporting Directive (CSRD). This directive expands reporting obligations to include small and medium-sized enterprises and prescribes the use of European Sustainability Reporting Standards, thereby seeking to standardize ESG disclosures across the region (Celli et al., 2024). Preliminary evidence suggests that the CSRD will improve the comparability and reliability of ESG data (Aluchna et al., 2023; Cicchiello et al., 2023), although there are concerns

about whether smaller firms have sufficient resources and expertise to comply with the stricter requirements (Ferrer et al., 2020; Lee et al., 2024).

Overall, the EU's transition from the NFRD to the CSRD reflects a broader trend toward more centralized ESG oversight. At the same time, it raises questions about how to balance the desire for comprehensive ESG disclosures with the administrative and financial burdens placed on organizations, particularly those with limited resources (Radu et al., 2023; Venturelli et al., 2022). This tension underscores the ongoing debate about ensuring both robustness and feasibility in ESG reporting standards throughout Europe.

United States ESG Legislation

In analyzing the recent history of ESG regulation in the United States, a complex and often contradictory landscape emerges, characterized by significant tensions between federal and state-level initiatives.

Anti-ESG Context

Anti-ESG efforts within federal and state congresses perceive ESG integration in financial markets as a threat to the federal and state economies (Jordan et al., 2023). At face value this seems counter to growing body of research which has examined the impact of ESG on firm financial performance in resource-intensive industries (Naeem et al., 2022; Alvarez-Perez & Fuentes, 2024). For instance, Alvarez-Perez and Fuentes (2024) found that ESG disclosures by oil and gas firms signal environmental stewardship and regulatory compliance, ultimately reducing the cost of debt by mitigating lenders' concerns over investment risks in the event of an environmental crisis. This finding aligns with earlier work by Naeem et al. (2022), which demonstrated that environmentally sensitive firms with higher ESG scores tend to be more

profitable and command higher market valuations. Conversely, firms with poor environmental performance face higher credit spreads (Zhang et al., 2022).

Despite these potential long-term financial benefits of ESG integration, anti-ESG legislation continues to proliferate at both federal and state levels. Laws such as Oklahoma's Energy Discrimination Elimination Act and Utah's Executive Order Review Process Amendments are frequently presented as necessary responses to what legislators view as federal overreach and challenges to state autonomy (Executive Order Review Process Amendments, 2021; Energy Discrimination Elimination Act, 2022). These initiatives reflect a political narrative in which ESG policies are construed as threats to state-controlled economic policies, particularly in resource-dependent regions. The short-term economic incentives behind these legislative actions are evident in research by Tang et al. (2024), which found that when state governments passed anti-ESG legislation, environmentally sensitive firms experienced positive abnormal stock returns, with even greater gains for firms with a history of poor environmental performance. This suggests that the deregulatory stance of anti-ESG legislators prioritizes immediate economic gains over the longer-term benefits identified in broader ESG research.

Anti-ESG Legislation

State-level anti-ESG legislation has proliferated in response to perceived economic and political threats posed by ESG, with lawmakers implementing various strategies to curtail its influence. Texas emerged as an early leader in this movement with the passage of Senate Bill 13 (Texas SB 13, 2021), which mandated a blacklist of financial institutions that boycott the oil, gas, and firearms industries and prohibited state contracts with such entities. This approach reflects a broader effort to insulate key industries from ESG-related financial exclusions. *House Bill 3* in Florida expanded restrictions further, prohibiting government entities from engaging

with companies that incorporate ESG factors in decision-making while mandating the use of only pecuniary factors in investment assessments (Government and Corporate Activism Act, 2023).

Anti-ESG sentiment has also extended into the insurance sector, as evidenced by Texas' *Senate Bill 833* and North Dakota's *House Bill 1429*, which prohibit insurers from adjusting rates or denying coverage based on ESG factors unless they are financially material (House Bill 1429, 2023; Senate Bill 833, 2023). These legislative efforts underscore the intensifying backlash against ESG investing in conservative-led states, reinforcing an economic paradigm that prioritizes short-term financial returns over long-term ESG risk-adjusted decisions that happen to align with sustainability considerations.

A recurring theme among these laws is the establishment of a “sole fiduciary” framework that mandates state investment decisions be based exclusively on pecuniary factors. For example, legislation in states such as Arkansas , Indiana , Georgia , and New Hampshire explicitly prohibits the consideration of nonfinancial ESG criteria in pension and public fund management (House Bill 457, 2023; Pension Investments, 2023; State Government Employee Retirement Protection Act, 2023; Public Retirement Systems Investment Authority Law, 2024). These bills are premised on the argument that ESG considerations introduce subjective, political, or ideological biases that could undermine the primary fiduciary duty to maximize financial returns for beneficiaries (Anti-ESG Act, 2023; Tennessee Investment Act, 2023).

Another prominent category of anti-ESG legislation targets public contracting practices. Several states have enacted measures requiring that companies contracting with state or local governments certify they do not engage in “economic boycotts” of specific industries such as fossil fuels, firearms, timber, and agriculture. For instance, bills in Alabama, Arkansas, and Utah

deploy certification mechanisms to ensure that state contracts are awarded only to firms that refrain from boycotting targeted industries (Anti-ESG Act, 2023; Prohibition on Public Entities From Contracting With Companies That Boycott Energy, Fossil Fuel, Firearms, and Ammunition Industries Act, 2023; Public Contract Requirements, 2023). This approach is framed as a defense of traditional economic sectors against what is perceived as discriminatory ESG-based practices (An Act to Amend the Law Concerning Environmental, Social Justice, or Governance Scores, 2023; Idaho Code, 2023).

Legislative actions also extend to the realm of financial services and lending. States such as Tennessee and New Hampshire have introduced statutes that expressly prohibit the use of “social credit scores” or any non-risk-based factors in banking and insurance decisions (Prohibiting Banks or Businesses From Using Social Credit Scores, 2022; Tennessee Investment Act, 2023). These provisions aim to prevent discrimination against customers based on their political, religious, or social attributes and to ensure that financial institutions operate within a strictly quantitative and risk-based framework. Similar measures can be seen in bills that restrict proxy voting practices in public pension funds, thereby ensuring that investment decisions are made exclusively for financial purposes (House Concurrent Resolution 110, 2023; West Virginia Investment Management Board and Board of Treasury Investments Shareholder Voting Requirements Act, 2023).

A further point of convergence is the broader political framing of these measures. Anti-ESG legislation is frequently presented as a necessary response to what legislators view as overreach by federal agencies and a challenge to state autonomy. Laws such as Oklahoma’s Energy Discrimination Elimination Act and Utah’s Executive Order Review Process Amendments underscore the intent to curtail federal mandates that incorporate ESG criteria into

public investment and regulatory practices (Executive Order Review Process Amendments, 2021; Energy Discrimination Elimination Act of 2022, 2022) . These initiatives reflect a political narrative in which ESG policies are construed as threats to state-controlled economic policies, particularly in resource-dependent regions.

Structural Problems with Anti-ESG Legislation

A particularly concerning aspect of some laws involves the transfer of proxy voting authority from pension fund trustees to elected officials. This contradicts empirical evidence suggesting that participant/beneficiary trustees typically outperform politicians in governance roles. Studies by Hess (2016) and Romano (1993) have demonstrated that pension funds with greater beneficiary representation tend to achieve better returns than those dominated by political appointees. This shift in governance structure may therefore compromise fund performance to the detriment of beneficiaries.

The economic consequences of anti-ESG legislation are beginning to emerge in several forms. Following Texas's implementation of anti-ESG laws, five large banks exited the Texas municipal bond market, resulting in an estimated \$300-500 million in additional borrowing costs (Garrett & Ivanov, 2022). A separate analysis by Econsult Solutions Inc. (2023) projected "up to \$700 million in higher municipal borrowing costs in six other states" implementing similar legislation. In Indiana, a fiscal impact statement estimated that anti-ESG legislation "could lower pension fund returns by up to \$7 billion over the next 10 years" (Legislative Services Agency, 2023). These concrete economic impacts suggest that anti-ESG laws may be imposing significant costs on taxpayers and beneficiaries while failing to achieve their stated objectives.

Pro-ESG Legislation

A prominent trend involves targeted divestment and climate risk disclosure. For example, California Senate Bill 1089 conditions the divestment of Turkish government securities on federal sanctions linked to historical human rights abuses, thereby intertwining geopolitical considerations with investment strategy (Public Employee Retirement Systems: Prohibited Investments: Turkey, 2022). Similarly, California’s SB 261 and SB 253 mandate comprehensive climate-related financial risk disclosures and greenhouse gas emissions reporting for large corporations, respectively, signaling a move toward enhanced transparency in environmental risk management (Climate Corporate Data Accountability Act, 2023; Senate Bill 261, 2023).

In the realm of social governance, legislative measures such as California Assembly Bill 979 and Senate Bill 655 focus on broadening board diversity by requiring representation from underrepresented communities and defining “diverse board member” in inclusive terms (Assembly Bill 979, 2020; Senate Bill No. 655, Chapter 390, 2021). Parallel efforts in Illinois through SB 1730 and SB0653 extend these principles by incorporating LGBTQ representation and aligning state investment practices with ESG criteria, thereby addressing systemic inequities in corporate leadership (Corporations LGBTQ Directors, 2021; Public Act 102-0297, 2021).

Several states have also leveraged ESG policy to address environmental sustainability directly. Colorado’s SB 16 sets ambitious greenhouse gas reduction targets across multiple sectors, while Maine’s fossil fuel divestment legislation and offshore wind energy initiatives combine divestment with proactive renewable energy development (An Act To Require the State To Divest Itself of Assets Invested in the Fossil Fuel Industry, 2021; An Act Regarding the Procurement of Energy from Offshore Wind Resources, 2023; Senate Bill 16, 2023). These

policies underscore an integrated strategy that uses state financial power to both mitigate climate risks and foster a transition to clean energy.

Further, the incorporation of ESG criteria into public investment and retirement systems is evident in Delaware's HB 334 (2020), Maryland's Climate Risk Fiduciary Duty measures (SB 566/HB 740, 2022), and Rhode Island's Secure Choice Retirement Savings Program Act (2024) (An Act To Amend Title 12 Of The Delaware Code Relating To Decedents Estates And Fiduciary Relations, 2020; State Retirement and Pension System - Investment Climate Risk - Fiduciary Duties, 2022; Rhode Island Secure Choice Retirement Savings Program Act, 2024). These policies institutionalize ESG factors in fiduciary practices, ensuring that investment decisions account for sustainable outcomes while safeguarding financial returns.

Collectively, these legislative initiatives highlight several key themes: the growing emphasis on mandatory environmental risk disclosures, a commitment to rectifying social inequities through enhanced diversity requirements, and the strategic use of state investments to drive sustainable and equitable economic outcomes. Despite variations in design and implementation across states, the overall trend suggests an emerging national framework where ESG considerations are increasingly recognized as vital to both financial stability and broader societal well-being. This synthesis not only underscores the breadth of current ESG policies but also points to the need for continued research on harmonizing these disparate initiatives to establish more unified, long-term sustainability standards.

The emerging tensions between pro- and anti-ESG policies highlight the intricate and polarized nature of America's ESG regulatory environment. Companies must navigate these differences by adapting their reporting practices to comply with evolving regulations while addressing investor demands for clearer ESG information (Arun et al., 2022). As legislative

debates continue, the trajectory of U.S. ESG policy will likely hinge on the delicate balance between market-driven sustainability expectations and efforts to shield specific industries from what some perceive as overly restrictive ESG criteria.

Firm Determinants of ESG Performance

Firm characteristics significantly influence corporate ESG outcomes. Understanding these determinants is essential to accurately assess ESG performance, as firm-specific traits influence strategic decisions, resource allocation, and reporting behaviors (Wald, 1999; Hendricks & Singhal, 2001; Solakoglu & Demir, 2016; Abideen, 2023). Unaccounted firm characteristics can confound analyses, particularly when evaluating policy impacts or corporate strategies (Memon, 2024).

Firm Size and Age

Among firm characteristics, size and age are consistently identified as critical determinants of ESG performance (Khaled et al., 2021; Lestari & Adhariani, 2022). Larger and more established firms typically possess greater resources, allowing substantial investment in sustainability practices and advanced technologies. Such financial flexibility often remains inaccessible to smaller or younger firms due to resource constraints (Ahmad et al., 2021; Kim & Li, 2021; Gholami et al., 2022). Consequently, overlooking these characteristics may yield misleading conclusions regarding ESG performance determinants, highlighting the need to control for firm size and age in empirical ESG research.

Financial Characteristics: Leverage, Profitability, and Market Valuation

Financial structure, particularly leverage, also plays a notable yet complex role in ESG performance. Khaled et al. (2021), analyzing 1202 firms across emerging markets, identified a positive relationship between leverage and ESG outcomes, explained through agency theory.

Firms with higher leverage may prioritize ESG reporting to demonstrate effective risk management to investors and creditors, thus reducing debt-associated risks and costs. However, this relationship may be context-dependent, particularly pronounced in emerging markets where agency costs are substantial, and transparency is critical (Chan et al., 2017).

Profitability indicators, specifically return on assets, exhibit a positive correlation with ESG performance. Firms with higher profitability generally possess better capacities to invest in sustainability initiatives, aligning financial strength with enhanced ESG outcomes (Miroshnychenko et al., 2017; Garcia & Orsato, 2020; Khaled et al., 2021). Similarly, firms exhibiting higher market valuations, measured by Tobin's Q, are more likely to integrate sustainability into strategic operations, reflecting a market-driven incentive to maintain strong ESG performance (Ortas et al., 2015).

Board Composition: Gender Diversity and Tenure

Board gender diversity is increasingly recognized as a significant determinant of ESG performance. Gender diverse boards are less prone to groupthink, enabling more comprehensive evaluations of ESG-related risks and opportunities (Cambrea et al., 2023). Historically, research linking Board gender diversity to firm performance primarily focused on financial outcomes (Post & Byron, 2015). Recent studies extend this focus to ESG performance, finding positive relationships between board gender diversity and ESG outcomes across various regions, including the United States, China, and Taiwan (Chang et al., 2024; Shakil, 2021; Xiaomei et al., 2021). However, the strength of this relationship varies geographically, influenced by regional gender parity norms and governance practices (AlJanadi, 2025; S. Wu et al., 2022). Thus, geographic context must be considered when evaluating the impact of board gender diversity on ESG outcomes.

Board tenure is another influential factor, with research presenting nuanced implications for ESG performance. According to the "expertise hypothesis," longer board tenure positively affects ESG performance by enhancing decision-making through accumulated knowledge and industry-specific insights (Schnake et al., 2011). Studies by Cremona and Passador (2019) and Andreassen and Bukhari (2024) confirm a positive correlation between tenure length and improved environmental and social outcomes. Conversely, extended board tenure is also associated with heightened risk aversion (Beasley, 1996; Katmon et al., 2019) and diminished oversight objectivity, potentially compromising ESG performance (Sharma & Iselin, 2012). This dual perspective highlights the complex relationship between board tenure and ESG, suggesting a balance between leveraging experience and preventing complacency is crucial.

Overall, firm characteristics significantly shape ESG outcomes, encompassing organizational attributes such as size and age, financial structures like leverage and profitability, and governance elements including board diversity and tenure. Recognizing and controlling these variables are vital for robust and insightful ESG performance analysis.

Regulatory Frameworks as Determinants of ESG Performance

Government regulations significantly influence corporate ESG performance, prompting diverse corporate responses depending on the nature and context of regulatory interventions. A broad range of governmental actions—including mandatory regulatory frameworks, targeted subsidies, taxation policies, transparency mandates, and public environmental investments—has been identified in the literature as critical determinants of ESG outcomes. While corporate reactions vary across regions and regulatory contexts, consistent themes emerge regarding the role of regulations in shaping corporate ESG strategies and performance.

Mandatory ESG Disclosure and Reporting Regulations

Mandatory ESG disclosure requirements have become widespread, particularly in Europe and Asia, as governments aim to enhance corporate transparency, accountability, and comparability of sustainability performance. Wu et al. (2025) documented significant improvements in ESG scores following the implementation of mandatory ESG disclosure rules by the Hong Kong Stock Exchange in 2016. The improvements were especially pronounced in firms within highly polluting industries, indicating the effectiveness of mandated disclosure in driving substantive corporate change. Similarly, Cicchiello et al. (2023) and Mbanyele et al. (2022) found consistent outcomes in Europe, where the introduction of the EU Non-Financial Reporting Directive (NFRD) significantly boosted corporate sustainability efforts and ESG performance. However, Arvidsson and Dumay (2022) highlighted a critical nuance, noting that while mandatory disclosures increased reporting volume substantially, they concurrently resulted in a plateau in disclosure quality. This stagnation emerged as firms increasingly adopted standardized, compliance-oriented behaviors rather than deeper strategic engagement with ESG issues.

These findings underscore an essential consideration for regulators: the distinction between disclosure quantity and quality, and how regulatory design impacts corporate ESG behavior in varying dimensions.

Strategic Adaptation to Regulatory Changes

Firms consistently employ strategic maneuvers to navigate regulatory uncertainty and evolving regulatory landscapes, strategically calibrating their compliance and ESG investments in response to anticipated regulatory actions and eventual enforcement intensity. Initially, companies may adopt proactive measures aimed at mitigating regulatory uncertainty and

establishing early compliance, followed by strategic recalibration upon actual enforcement. Cicchiello et al. (2023) further elaborated on this pattern within the European Union context, highlighting an initial minimal compliance approach by firms that evolves into deeper ESG engagement following more stringent enforcement actions.

A notable illustrative case from Garrett and Ivanov (2022) occurred in Texas, where legislation prohibited financial institutions from incorporating ESG criteria into bond evaluations. This compelled lending firms to respond strategically, either exiting the market entirely or returning only under conditions of strict adherence to the legislation. The resulting pattern of repeated bond market exits and reentries underscores the complexities and strategic considerations corporations face in responding to evolving regulatory frameworks.

In the European setting, similar strategic adaptations emerged in response to the NFRD. Arvidsson and Dumay (2022) observed that initially voluntary disclosures provided firms with a market advantage, but as mandatory requirements became standardized, firms shifted their strategic focus toward baseline compliance rather than deeper ESG innovation. Thus, regulatory frameworks not only shape immediate corporate ESG outcomes but also influence firms' broader strategic decisions, often reflecting anticipatory behavior in response to perceived regulatory risks or opportunities.

Expanding Research on Regulatory Influence

Current literature emphasizes various dimensions of governmental influence beyond direct disclosure requirements, including financial incentives and policy-specific interventions. Subsidies and public investments, particularly those targeting environmental innovation, significantly improve corporate ESG outcomes. Niu (2024) demonstrated that increased public spending on environmental protection positively correlates with enhanced national ESG

performance. Targeted subsidies, as explored by Peng et al. (2024), also significantly impact firms, particularly non-state firms and those in heavily polluting industries, reinforcing how targeted financial interventions can amplify sustainability efforts effectively.

Furthermore, research identifies nuanced impacts resulting from carbon regulation policies. Carbon pricing mechanisms, specifically emissions trading schemes (ETS), notably drive corporate emissions reductions and ESG improvements (Mittelstaedt, 2024). Yet these benefits often disproportionately accrue to larger firms capable of leveraging extensive resources, raising important questions regarding regulatory equity and effectiveness.

The literature suggests several key mechanisms through which government influence operates: (1) mandatory disclosure requirements create transparency and accountability; (2) financial incentives and penalties alter cost-benefit calculations for ESG investments; (3) policy uncertainty triggers strategic ESG positioning; and (4) technological transformation requirements drive innovation that enhances ESG performance. These findings provide a foundational basis for examining how emerging state-level ESG policies may impact corporate sustainability efforts in the U.S. and highlight the need for carefully designed regulatory frameworks that balance compliance requirements with incentives for substantive improvement in ESG outcomes.

Industry Classification

Industry classification standards play a crucial role in organizing and categorizing economic activities for statistical and analytical purposes. The evolution of industry classification systems has seen the development of various standards, each with its unique features and methodologies.

The Standard Industrial Classification (SIC) system, established in 1941, was the first widely adopted industry classification method, initially designed to categorize both

manufacturing and non-manufacturing industries (Pearce, 1957). The SIC system underwent its last major update in 1987, resulting in a four-tiered framework with 11 segments at the top tier and 1,005 distinct segments at the base level, including a section for unclassifiable entities (Phillips & Ormsby, 2016).

However, the SIC system has faced criticism for its lack of guiding principles in its development (Phillips & Ormsby, 2016). To overcome the deficiencies identified in the SIC system, the collaborative endeavor of Mexico's, Canada's, and the United States statistical bodies led to the formulation of the North American Industry Classification System (NAICS) in 1998 (Phillips & Ormsby, 2016). The NAICS system aimed to provide a more comprehensive and methodologically sound approach to industry classification by segmenting industries based on their production methods. Unlike the SIC system, NAICS features five levels in the industry hierarchy, allowing for a more detailed and nuanced classification of economic activities (Phillips & Ormsby, 2016).

CHAPTER III

DATA

This study employs a comprehensive dataset covering publicly traded firms from 2019 to 2024. The data combines firm-level ESG performance metrics, financial characteristics, and state-level ESG legislation information to examine the relationship between regulatory pressures and corporate sustainability practices.

ESG Data

ESG performance data are sourced from S&P Global, which quantify how companies address key ESG factors. Each score ranges from 0 to 100 and reflects both a company's disclosures and its actual performance on sustainability issues. The scores are derived from the annual S&P Global Corporate Sustainability Assessment (CSA), which evaluates companies through approximately 130 industry-specific questions, supported by up to 1,000 data points. The CSA scoring frameworks award weighted points based on three dimensions: preparedness (quality of policies and programs), transparency (disclosure of qualitative or quantitative information), and performance (assessment of quantitative metrics against benchmarks).

For non-participating companies, S&P Global analysts utilize publicly available information to generate scores. Points earned at the question level are aggregated to broader criteria, then to the ESG dimensions, and finally to a single ESG score. This industry-specific approach implements tailored weighting schemes, assigning greater importance to criteria deemed most relevant for each sub-industry. S&P Global's (2024b, p. 11) methodology suggests analysts use the following model specifications (see Table 1) to calculate the ESG indices:

Table 1. S&P Global ESG Indices Calculations

Variable	Formula	Source
ESG Index Score	$\sum((SPQP \times SPQW) \times SPCW) \times SPDW$	S&P Global
Social Index Score	$\sum((SPQP_s \times SPQW_s) \times SPCW_s) \times SPDW_s$	S&P Global
Environmental Index Score	$\sum((SPQP_e \times SPQW_e) \times SPCW_e) \times SPDW_e$	S&P Global

Note: *SPQP* represents question-level performance, *SPQW* denotes question weights, *SPCW* indicates criteria weights, and *SPDW* signifies dimension weights. Personal communications with S&P Global’s Senior Management reveal these weights are proprietary to S&P Global.

Board Data

Board Gender Diversity and Board Tenure are S&P analyst calculated weighted indices in response to answers on Questions 1.2.5 and 1.2.7 in the S&P Global CSA questionnaire (S&P Global, 2024a, pp. 16–19). S&P Global (2024a) suggests their approach evaluates the actual representation of women rather than merely rewarding disclosure, reflecting research findings correlating gender diversity with improved corporate governance and innovation (Adams & Ferreira, 2009; Dezsö & Ross, 2012). S&P Global (2024a) further draws on research indicating optimal board tenure falls in the 7–12-year range (Canavan et al., 2004; S. Huang & Hilary, 2018), with firm value declining as average tenure deviates from this range. S&P analysts use the following model specification (see Table 2) to calculate the indices:

Table 2. S&P Global Gender Diversity and Tenure Indices Calculations

VARIABLE	Formula	Source
GENDER DIVERSITY	$\frac{\# \text{ Women}}{\# \text{ Board Members}} \times SPQW$	S&P Global
TENURE	$\frac{\text{Current Year} - \text{Appointment Year}}{\text{Number of Directors}} \times SPQW$	S&P Global

Note: *SPQW* denotes question weights. Both these Indices are components of the broader governance dimension score and are weighted according to industry-specific materiality assessments, reflecting their importance in corporate decision-making and long-term shareholder value.

Firm Data

Firm-specific financial characteristics are drawn from the Standard & Poor's (S&P) Compustat Annual database for the years 2019-2024. Compustat provides standardized accounting information for publicly listed companies, including key balance sheet, income statement, and cash flow items.

From this dataset, I construct several control variables (see table 3). Leverage, calculated as $\frac{DLTT + DLC}{SEQ}$, where $DLTT$ represents long-term debt, DLC denotes debt in current liabilities, and SEQ indicates shareholders' equity. Company Size, measured as the natural logarithm of total assets: $\log(AT)$. Return on Assets (ROA), computed as $\frac{NI}{AT}$, where NI represents net income and AT denotes total assets. Tobin's Q, calculated as $\frac{AT + (CSHO \times PRCC_F) - CEQ}{AT}$, where $CSHO$ represents common shares outstanding, $PRCC_F$ denotes fiscal-year closing share price, CEQ indicates common equity, and AT is total assets.

Table 3. Control Variable Calculations.

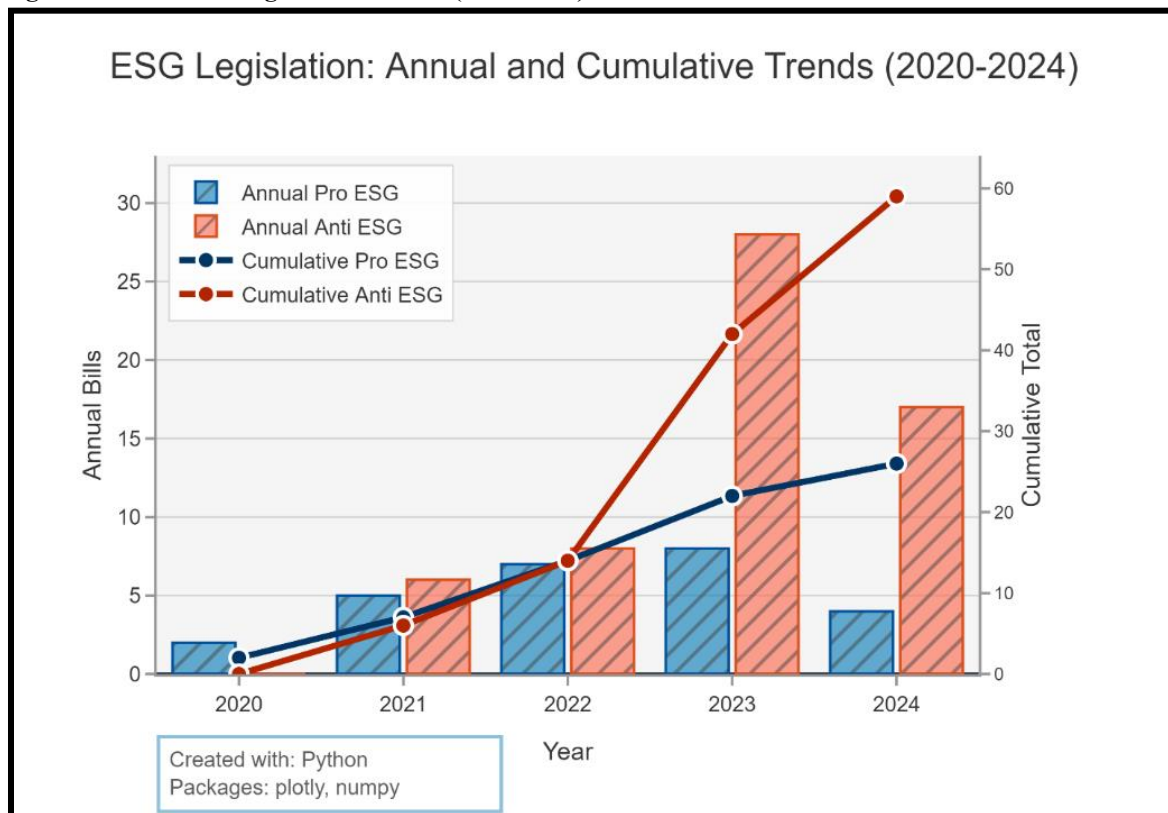
Variable	Formula	Source
Leverage	$\frac{DLTT + DLC}{SEQ}$	Compustat
Company Size	$\log(AT)$	Compustat
Return on Assets	$\frac{NI}{AT}$	Compustat
Tobin's Q	$\frac{AT + (CSHO \times PRCC_F) - CEQ}{AT}$	Compustat
Firm Age	$Observation\ Year - Year\ Founded$	S&P Global

Note: Control variables are calculated using Compustat data except for Firm Age (S&P Global). All financial variables were winsorized at 1% and 99% levels to mitigate outliers. Firms with negative leverage and ROA values were excluded as these typically represent distressed companies with negative equity, potentially distorting financial relationship interpretations.

Legislative Data

State-level ESG legislation data for this study were compiled from two comprehensive sources: LegiScan and Ballotpedia. Ballotpedia's ESG Tracker (2024) webpage provides a collection of links to all ESG bills enacted between 2020 and 2024, while LegiScan offers access to the complete corpus of U.S. legislation in the form of PDFs and text files. The methodology section will provide detailed information on the techniques employed to integrate these two data sources. Through this integration process, a total of 85 ESG legislative bills were identified for analysis. The temporal distribution of these bills reveals interesting patterns in ESG legislative activity (see figure 1).

Figure 1. State ESG Legislation Trends (2020-2024)



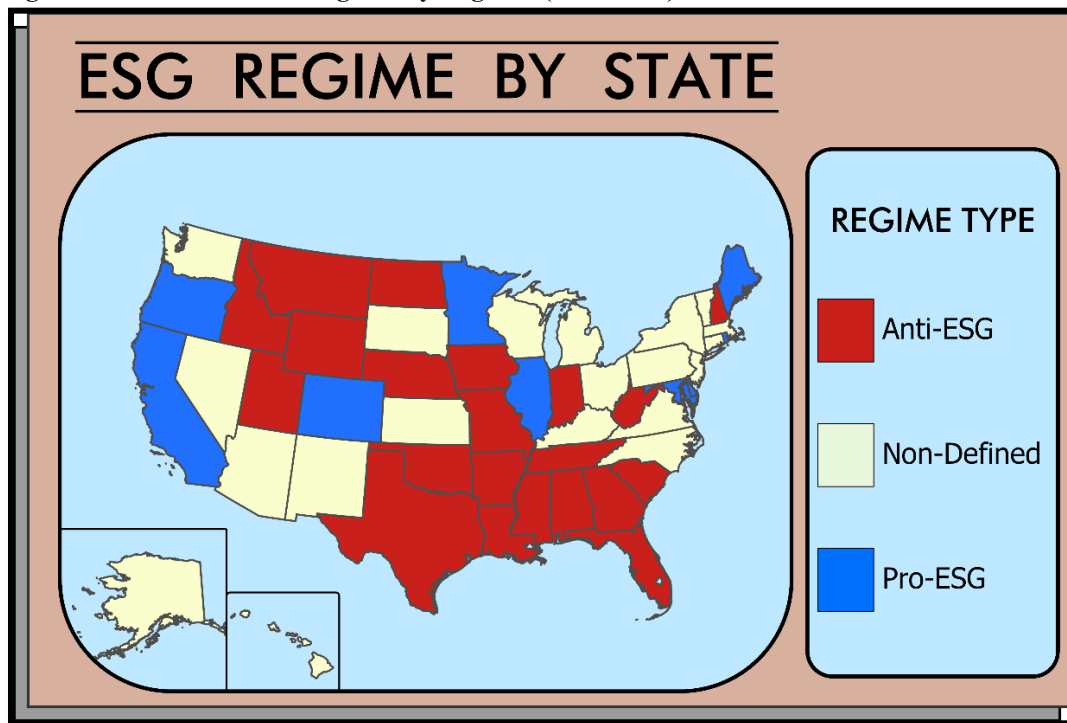
Note: Data compiled from LegiScan and Ballotpedia's ESG Tracker (2020-2024). Bills were classified as 'pro-ESG' if they mandated or encouraged consideration of environmental, social, or governance factors in investment decisions or corporate operations, and 'anti-ESG' if they explicitly restricted or prohibited such considerations.

From 2020 to 2024, I observed a clear evolution in both pro-ESG and anti-ESG legislation. Pro-ESG bills showed modest but steady growth from 2 bills in 2020 to 8 bills in 2023, followed by a slight decrease to 4 bills in 2024. In contrast, anti-ESG legislation exhibited a dramatic increase, starting from zero bills in 2020 and rising sharply to 28 bills in 2023, before declining somewhat to 17 bills in 2024.

This visualization illustrates the stark contrast between pro-ESG and anti-ESG legislative activity across the five-year period. The data reveal a notable inflection point in 2021-2022, when anti-ESG legislation began to gain significant momentum, ultimately outpacing pro-ESG legislation by a considerable margin in 2023. This trend suggests a growing politicization of ESG policies at the state level, with anti-ESG sentiment becoming increasingly prevalent in state legislatures.

The geographic distribution of ESG legislation (Figure 2) provides further evidence of this politicization. The map illustrates a clear regional pattern in ESG policy adoption, with anti-ESG legislation concentrated in the Southern, Midwestern, and Mountain West regions. States such as Texas, Florida, Oklahoma, Missouri, and others across these regions have enacted laws that explicitly restrict or oppose ESG considerations in investment decisions, particularly targeting state pension funds and government contracts.

Figure 2. State-Level ESG Regulatory Regimes (2020-2024)



Note: States are classified based on their predominant ESG legislative approach (2020-2024). 'Non-Defined' states have either not enacted significant legislation or have pursued a bipartisan approach outside the focus of this study.

Sample Construction

This study draws on a comprehensive panel dataset of publicly traded firms observed between 2019 and 2024. The sampling strategy targets companies operating in sectors directly exposed to state-level ESG legislation during this period. By concentrating on firms subject to varying degrees of regulatory intervention, the analysis can more precisely isolate the impact of policy shifts on corporate sustainability performance. The final dataset comprises 12,938 firm-year observations from 2,938 unique companies.

Table 4 presents key summary statistics. The average overall ESG score is 32.13 (SD = 13.42), indicating considerable variation in sustainability outcomes. Environmental and social dimension scores have similar mean scores of 28.45 and 26.98, suggesting that most firm practices vary across both environmental and social domains. In terms of regulatory environments, 4.9% of firm-year observations face anti-ESG policies, while 10.4% operate under

pro-ESG frameworks. This circumstance provides a quasi-natural experiment setting to assess how contrasting policy contexts influence firm ESG behaviors.

Table 4. Summary Statistics

Variables	Mean	Std. dev.	Min	Max
Overall ESG Score	32.127	13.415	6	91
Environmental Score	28.447	17.844	0	99
Social Score	26.982	14.128	0	92
Anti ESG Policy (1=Yes)	0.049	0.215	0	1
Pro ESG Policy (1=Yes)	0.104	0.305	0	1
Board Gender Diversity	51.017	30.748	0	100
Board Average Tenure	78.276	28.441	0	100
Company Age	61.230	48.110	1	251
Company Size	8.388	1.769	3.128	15.203
Leverage	1.418	2.396	0.0000164	16.244
Return on Assets	0.015	0.132	-0.998	0.393
Tobin's Q	2.186	2.026	0.490	27.896

Note: This table presents summary statistics for the key variables in the analysis.

Governance-related metrics also exhibit wide dispersion. Board gender diversity, for example, averages 51.02% but has a substantial standard deviation of 30.75%, highlighting marked cross-sectional differences. Similarly, the average board tenure is 78.28%, reflecting generally seasoned leadership. Firm demographics vary greatly: the average company age is 61.23 years, while company size (log of total assets) averages 8.39, indicating notable heterogeneity in scale, and leverage averages 1.42.

Turning to performance indicators, the mean return on assets stands at 1.5%, whereas the mean Tobin's Q is 2.19. These figures suggest that, on balance, the market values sampled firms above their replacement costs. However, both performance measures have considerable amount of variance, reflecting the substantial diversity in financial outcomes across these companies.

CHAPTER VI

JOURNAL ARTICLE

Introduction

Corporate philosophy has evolved dramatically since Adam Smith's foundational work *The Wealth of Nations* (1776), which established that free market competition drives economic prosperity through the "invisible hand" mechanism. This classical perspective was later reinforced by Milton Friedman (1970), who argued that a business's sole social responsibility is profit maximization within legal boundaries. However, this paradigm has faced significant challenges as sustainability concerns have entered the mainstream of business discourse.

The emergence of CSR was catalyzed by high-profile incidents like the 1990s Nike labor scandal, where the revelation of exploitative labor practices in developing countries damaged the company's reputation and financial performance (Ballinger, 1992; Schanberg, 1996). Nike's subsequent establishment of accountability measures and transparency initiatives demonstrated the growing importance of social responsibility as a business imperative rather than merely a moral consideration (Murphy & Mathew, 2001; Achabou, 2020).

The institutionalization of ESG factors gained momentum through the United Nation's 2004 "Who Cares Wins" report that provided a framework for integrating ESG factors into investment analysis and asset management (UN Environment Programme – Finance Initiative, 2004). This framework has evolved into a global industry of ESG data providers and metrics, despite ongoing challenges with standardization and measurement validity (Eccles et al., 2020).

While regions like the European Union and China have increasingly embraced ESG through regulatory mandates (European Parliament, 2022; Ministry of Ecology and Environment, 2021), the United States presents a more complex landscape. A significant

countermovement has emerged along political party lines, with states enacting conflicting pro-ESG and anti-ESG legislation. Texas's anti-ESG Senate Bills 13 and 19, for instance, have excluded major financial institutions from government contracts, resulting in approximately \$400 million in increased annual borrowing costs for taxpayers (Texas Senate Bill 13, 2021; Texas Senate Bill 19, 2021; Garrett & Ivanov, 2022).

This study expands upon existing research by examining how both pro-ESG and anti-ESG state legislation impacts corporate ESG performance across multiple jurisdictions. While economic analyses like Garrett and Ivanov's (2022) have quantified immediate financial consequences, my research provides a more comprehensive understanding of how these divergent policy approaches affect corporate behavior and sustainability outcomes. Through empirical analysis of state-level ESG policies and their relationship to corporate ESG metrics, this study contributes to the ongoing debate about the role of government in promoting or restricting sustainable business practices in the American federalist system.

The remainder of this journal ready article is structured as follows. Section 2 will review relevant literature. Section 3 will discuss data collection. Section 4 elaborates on the empirical method. Section 5 will present a discussion of results and findings.

Literature Review

This section reviews the literature on ESG theory, firm determinants of ESG performance, and ESG legislation in the European Union and United States. It begins with an overview of ESG data development, followed by an examination of the theoretical foundations of ESG. Next, it analyzes EU and U.S. ESG regulatory frameworks, with particular attention to their divergent approaches. The section concludes with an exploration of firm-level characteristics that drive ESG performance.

ESG Data: Evolution and Challenges

ESG metrics have evolved from nascent corporate social responsibility (CSR) assessments into sophisticated data products essential to modern investment analysis. Initially reliant on company self-reporting, ESG evaluation methodologies now incorporate diverse data sources including regulatory filings, news analytics, and social media sentiment (De Giuli et al., 2024). This evolution has been accelerated by technological advancements in big data analytics and artificial intelligence, enhancing the efficiency and reliability of ESG assessments (Saxena et al., 2022).

Despite these improvements, significant measurement challenges persist. Berg et al. (2022) documented substantial divergence between ESG ratings agencies evaluating the same companies, attributable to variations in measurement methodologies, categorical scope, and weighting schemes. These inconsistencies create ambiguity for investors and companies alike, reflecting the complex and context-dependent nature of ESG evaluation (Chatterji et al., 2016). Additionally, the regulatory environment has enabled greenwashing practices, where firms exaggerate their ESG commitments to enhance market perceptions without substantive action (Chen et al., 2023).

Theoretical Foundations of ESG

Signaling theory provides a critical framework for understanding ESG disclosures as strategic corporate communications intended to reduce information asymmetry between firms and stakeholders. Developed by Spence (1973, 2002), this theory posits that organizations signal their quality and intentions through observable actions when information asymmetry exists between information holders (firms) and receivers (stakeholders). In ESG contexts, effective signaling can enhance stakeholder trust and improve market positioning.

Research demonstrates that capital markets respond positively to ESG signals, particularly when firms integrate financial and ESG disclosures into unified reports (Mervelskemper & Streit, 2017). Importantly, the effectiveness of ESG signals depends on both their novelty and quality; firms with no prior history of ESG disclosures experience significantly higher market valuations upon their first ESG report (Wang & Li, 2016). However, routine disclosures may diminish signaling effectiveness if they fail to introduce new information that differentiates the company from competitors (Connelly et al., 2011; Xie et al., 2019).

Critical Mass Theory offers insights into how board diversity affects ESG outcomes. Originally developed by Granovetter (1978) and Olson (1965) and later applied to corporate governance by Kanter (1977), this theory suggests that a minimum representation—typically three or more individuals from underrepresented groups—is necessary to effect meaningful change in group dynamics and decision-making.

Empirical research strongly supports this theoretical framework in ESG contexts. Konrad et al. (2008) demonstrated that boards with three or more women exhibited improved governance and decision-making effectiveness. Yarram and Adapa (2021) provided further evidence in their study of ASX 300 companies, showing that minimal female representation had no significant impact on corporate social responsibility, likely due to tokenism challenges. However, when companies achieved a critical mass of three or more women directors, CSR performance significantly improved, with an effect size of 5.771 points ($p < 0.05$).

European Union ESG Legislation: Progressive Standardization

The 2014 Non-Financial Reporting Directive (NFRD) marked the EU's first major step toward mandatory ESG disclosure, requiring large public-interest entities to report on environmental issues, social matters, human rights, anti-corruption practices, and diversity policies (Arif et al., 2022; Lombardi et al., 2022). Empirical evidence indicates this mandate substantially increased non-financial disclosure volume, particularly among firms with limited prior ESG reporting (Aluchna et al., 2023; Čufar & Belak, 2024). However, inconsistent compliance across member states highlighted the limitations of the NFRD's flexible implementation approach (Cicchello et al., 2023; Hao et al., 2023).

Addressing these shortcomings, the Corporate Sustainability Reporting Directive (CSRD) expanded both the scope and standardization of ESG reporting. The CSRD extends disclosure obligations to small and medium-sized enterprises while prescribing European Sustainability Reporting Standards (ESRS) to enhance comparability (Celli et al., 2024). Early assessments suggest the CSRD will significantly improve ESG data reliability (Aluchna et al., 2023; Cicchello et al., 2023), though concerns remain about resource burdens on smaller firms (Ferrer et al., 2020; Lee et al., 2024).

The EU's transition from the NFRD to the CSRD exemplifies a deliberate progression toward more centralized and comprehensive ESG oversight, balancing disclosure requirements against implementation feasibility (Radu et al., 2023; Venturelli et al., 2022). This standardization approach contrasts sharply with the polarized regulatory landscape in the United States.

United States ESG Legislation: Polarization and Fragmentation

In contrast to the EU's relatively unified approach, the United States presents a fragmented and increasingly polarized ESG regulatory landscape characterized by tensions between federal initiatives and state-level resistance. This fragmentation creates significant compliance challenges for corporations operating across multiple jurisdictions.

Anti-ESG State Legislation.

State-level anti-ESG legislation has proliferated based on the perception that ESG integration threatens state economies, particularly in resource-dependent regions. Texas pioneered this movement with Senate Bill 13, which created blacklists of financial institutions deemed to boycott oil, gas, and firearms industries and prohibited state contracts with such entities (Texas SB 13, 2021). These laws produced significant economic consequences: five major banks exited the Texas municipal bond market, resulting in estimated additional borrowing costs of \$300-500 million (Garrett & Ivanov, 2022).

Several common approaches characterize anti-ESG legislation across states. First, many states have established "sole fiduciary" frameworks mandating that state investment decisions be based exclusively on pecuniary factors. Legislation in Arkansas, Indiana, Georgia, and New Hampshire explicitly prohibits consideration of nonfinancial ESG criteria in pension and public fund management (House Bill 457, 2023; Public Employee Retirement Systems: Prohibited Investments: Turkey, 2022; State Government Employee Retirement Protection Act, 2023; State Retirement and Pension System - Investment Climate Risk - Fiduciary Duties, 2022).

Second, many states have targeted public contracting practices by requiring that companies contracting with state governments certify they do not engage in "economic boycotts" of specific industries. Bills in Alabama, Arkansas, and Utah deploy certification mechanisms to

ensure state contracts are awarded only to firms that refrain from boycotting targeted industries (Anti-ESG Act, 2023; Prohibition on Public Entities From Contracting With Companies That Boycott Energy, Fossil Fuel, Firearms, and Ammunition Industries Act, 2023; Public Contract Requirements, 2023).

Third, some states have restricted financial services practices. Tennessee and New Hampshire have prohibited the use of "social credit scores" or non-risk-based factors in banking and insurance decisions (Prohibiting Banks or Businesses From Using Social Credit Scores, 2022; Tennessee Investment Act, 2023). These provisions aim to prevent discrimination based on political, religious, or social attributes while ensuring strict adherence to quantitative risk assessment frameworks.

Despite the economic justifications offered for anti-ESG legislation, emerging evidence suggests these laws may impose significant costs. Econsult Solutions Inc. (2023) projected "up to \$700 million in higher municipal borrowing costs in six other states" implementing legislation similar to Texas. In Indiana, a fiscal impact statement estimated that anti-ESG legislation "could lower pension fund returns by up to \$7 billion over the next 10 years" (Legislative Services Agency, 2023).

Interestingly, this political opposition contrasts with research demonstrating financial benefits of ESG integration in resource-intensive industries. Alvarez-Perez and Fuentes (2024) found that ESG disclosures by oil and gas firms signal environmental stewardship and regulatory compliance, ultimately reducing debt costs by mitigating lenders' concerns about environmental risk. Similarly, Naeem et al. (2022) demonstrated that environmentally sensitive firms with higher ESG scores tend to be more profitable and command higher market valuations after the passing of anti-ESG legislation.

Pro-ESG State Legislation.

Concurrently, several states have implemented pro-ESG legislation focused on disclosure requirements, board diversity, and sustainable investment. California has been particularly active, with SB 261 and SB 253 mandating comprehensive climate-related financial risk disclosures and greenhouse gas emissions reporting for large corporations (Climate Corporate Data Accountability Act, 2023; Senate Bill 261, 2023). Other states have leveraged ESG policy to address environmental sustainability directly. Colorado sets ambitious greenhouse gas reduction targets across multiple sectors, while Maine's fossil fuel divestment legislation combines divestment with proactive renewable energy development (An Act To Require the State To Divest Itself of Assets Invested in the Fossil Fuel Industry, 2021; Senate Bill 16, 2023). Additionally, Delaware, Maryland, and Rhode Island have institutionalized ESG factors in fiduciary practices for public investments and retirement systems (An Act To Amend Title 12 Of The Delaware Code Relating To Decedents Estates And Fiduciary Relations, 2020; Rhode Island Secure Choice Retirement Savings Program Act, 2024; State Retirement and Pension System - Investment Climate Risk - Fiduciary Duties, 2022). This bifurcated regulatory landscape creates significant challenges for corporations operating across multiple states, requiring nuanced compliance strategies tailored to each jurisdiction's requirements and restrictions.

Firm Determinants of ESG Performance

Corporate ESG performance is significantly influenced by firm-specific characteristics that shape strategic decisions, resource allocation, and reporting behaviors. Understanding these determinants is essential for accurate assessment of ESG outcomes and effective policy design (Abideen, 2023; Hendricks & Singhal, 2001; Solakoglu & Demir, 2016).

Organizational Characteristics: Size and Age.

Firm size and age consistently emerge as critical determinants of ESG performance (Khaled et al., 2021; Lestari & Adhariani, 2022). Larger, more established firms typically possess greater resources for sustainability investments and advanced technologies that remain inaccessible to smaller or younger counterparts (Ahmad et al., 2021; Gholami et al., 2022). This resource advantage translates into superior ESG outcomes across ESG dimensions. Consequently, analyses that fail to control for these organizational characteristics risk misattributing ESG performance effects to other factors (Wang et al., 2022).

Financial Structure and Performance.

Financial characteristics, particularly leverage and profitability, significantly influence ESG performance. Khaled et al. (2024) identified a positive relationship between leverage and ESG outcomes across 1,202 firms in emerging markets, explained through agency theory. Highly leveraged firms often prioritize ESG reporting to demonstrate effective risk management to creditors, thereby reducing debt-associated costs. However, this relationship appears context-dependent, more pronounced in markets where transparency is critical to investor confidence (Chan et al., 2017).

Profitability indicators, specifically Return on Assets (ROA), consistently correlate positively with ESG performance. Firms with higher profitability generally possess enhanced capacity to invest in sustainability initiatives, creating a virtuous cycle between financial strength and ESG outcomes (Garcia & Orsato, 2020; Miroshnychenko et al., 2017). Similarly, firms with higher market valuations (measured by Tobin's Q) demonstrate stronger ESG performance, reflecting market-driven incentives for sustainability investment (Ortas et al., 2015).

Board Composition: Gender Diversity and Tenure.

Board gender diversity increasingly emerges as a significant determinant of ESG performance. Gender-diverse boards exhibit less groupthink, enabling more comprehensive evaluation of ESG-related risks and opportunities (Cambrea et al., 2023). Recent studies demonstrate positive relationships between board gender diversity and ESG outcomes across various regions, including the United States, China, and Taiwan (Chang et al., 2024; Shakil et al., 2021; Xiaomei et al., 2021). However, relationship strength varies geographically, influenced by regional gender parity norms and governance practices (AlJanadi, 2025; Wu et al., 2022).

Board tenure presents more complex implications for ESG performance. Studies by Cremona and Passador (2019) and Andreassen and Bukhari (2024) confirm positive correlations between tenure length and improved environmental and social outcomes. Conversely, extended tenure is also associated with heightened risk aversion (Beasley, 1996; Katmon et al., 2019), increased informal industry connections, and diminished oversight objectivity (Sharma & Iselin, 2012). This duality suggests optimal ESG performance requires balancing experience utilization against preventing governance complacency.

Data

This study examines the relationship between state-level ESG legislation and corporate sustainability practices using a comprehensive dataset of publicly traded U.S. firms from 2019 to 2024. The final sample comprises 12,938 firm-year observations from 2,938 unique companies operating in sectors directly affected by ESG regulatory interventions.

ESG Data

This study employs S&P Global's Corporate Sustainability Assessment (CSA) data for firm-level ESG performance. S&P Global calculates these scores using approximately 130 industry-specific questions supported by up to 1,000 data points. The CSA scoring framework awards weighted points based on three dimensions: preparedness (policy quality), transparency (disclosure), and performance (quantitative metrics). For non-participating companies, S&P Global analysts generate scores using publicly available information. S&P Global calculates the ESG indices using the following specifications:

$$Y_{it} = \sum ((SPQT \times SPQW) \times SPCW) \times SPDW \quad (1)$$

Where Y_{it} is the environmental index score, social index score, or overall index score of a firm. $SPQP$ represents question-level performance, $SPQW$ denotes question weights, $SPCW$ indicates criteria weights, and $SPDW$ signifies dimension weights. These proprietary weights are tailored by industry, assigning greater importance to criteria deemed most relevant for each sub-industry. Per communications with S&P, the weighting information is not accessible for data subscribers such as me.

Next, this study collects board gender diversity and board tenure data from S&P global. As part of the CSA questionnaire that S&P administers annually, firm are asked via questions 1.2.5 and 1.2.7 what the firms board gender diversity and average tenure is (S&P Global, 2024a, pp. 16–19). Like the ESG index scores, S&P Global weights the gender diversity and tenure information using subjective measures of preparedness, transparency, and performance. S&P Global's analysts calculate these metrics through the following formulas:

$$\text{Board Gender Diversity Index Score} = \frac{\# \text{ Women}}{\# \text{ Board Members}} \times SPQW \quad (2)$$

$$\text{Board Tenure Index Score} = \frac{\text{Current Year} - \text{Appointment Year}}{\text{Number of Directors}} \times SPQW \quad (3)$$

Where SPQW represents S&P's quality weighting factor that accounts for the subjective assessment measures.

The ESG dataset was cleaned to ensure the integrity and relevance of the data included in the final analysis. Initial data cleaning involved filtering out any observations where the ESG score type was categorized as "Raw," as these values represent unprocessed figures that could introduce noise into the analysis. Additionally, entries with missing unique firm identifiers were removed to maintain consistency when merging datasets. The dataset was then refined to include only key variables such as institution identification, firm gvkey, firm name, assessment year, country and state of incorporation, and firm founding year. Any non-numeric values were coerced into missing values to prevent computational errors. Instances of controversy can trigger a CSA reassessment; thus, some firms have multiple ESG reporting for a given year. The approach of this study was to average out index scores where firms have index values in a given year due to reassessment.

Firm Data

To isolate the effects of my independent variables on ESG performance, I include a comprehensive set of firm-level control variables that prior literature has established as significant determinants of corporate sustainability practices (see table 3).

Firm Size is measured as the natural logarithm of total assets, consistent with established practice in the corporate sustainability literature (Khaled et al., 2021; Lestari & Adhariani, 2022). This logarithmic transformation addresses the right-skewed distribution of firm asset values and ensures the variable meets regression assumptions. The literature consistently indicates that larger firms demonstrate superior ESG performance due to greater resource availability,

heightened public scrutiny, and economies of scale in implementing sustainability initiatives (Ahmad et al., 2021; Wang et al., 2022).

Firm Age is calculated as the number of years since a firm's incorporation date until the observation year. This approach aligns with methodologies employed by Gholami et al. (2022) and Kim & Li (2021). More established firms typically exhibit more developed sustainability frameworks, as they have had sufficient time to institutionalize ESG practices and respond to evolving stakeholder pressures (Shu & Tan, 2023).

Leverage is operationalized as the ratio of total debt (sum of long-term debt and short-term debt) to total shareholders' equity. This measure captures a firm's financial risk exposure and capital structure decisions. The relationship between leverage and ESG performance exhibits contextual variation in the literature: while Khaled et al. (2024) documented a positive relationship in emerging markets attributed to agency theory mechanisms, others have identified negative associations with certain ESG dimensions (Barnea & Rubin, 2010; Yan et al., 2022). Environmental performance specifically tends to demonstrate a positive relationship with leverage (Sulaiman et al., 2014; Clarkson et al., 2008), potentially reflecting strategic legitimacy-seeking behavior by firms with higher debt levels.

Return on assets is measured as net income divided by total assets, representing a firm's efficiency in utilizing its asset base to generate earnings. This widely applied metric (Garcia & Orsato, 2020; Miroshnychenko et al., 2017) captures a firm's financial capacity to invest in sustainability initiatives, with the literature consistently demonstrating that more profitable firms exhibit superior ESG performance (Khaled et al., 2021).

Chung & Pruitt's (1994) outline the calculation of Tobin's Q as $(\text{Market Value of Equity} + \text{Book Value of Debt}) / \text{Book Value of Total Assets}$. This measure captures market perceptions

of a firm's growth prospects and intangible value. Firms with higher Tobin's Q ratios tend to demonstrate stronger ESG performance (Ortas et al., 2015), reflecting market recognition of sustainability initiatives as value-enhancing investments rather than discretionary expenditures.

To further refine the dataset, extreme outliers were winsorized at the 1st and 99th percentiles. This process ensured that extreme observations did not unduly influence statistical analyses. Additionally, infinite values resulting from division errors were replaced with missing values to maintain data validity. Firms with negative leverage values were excluded as these typically represent distressed companies with negative equity, which could distort financial relationship interpretations (see Table 3).

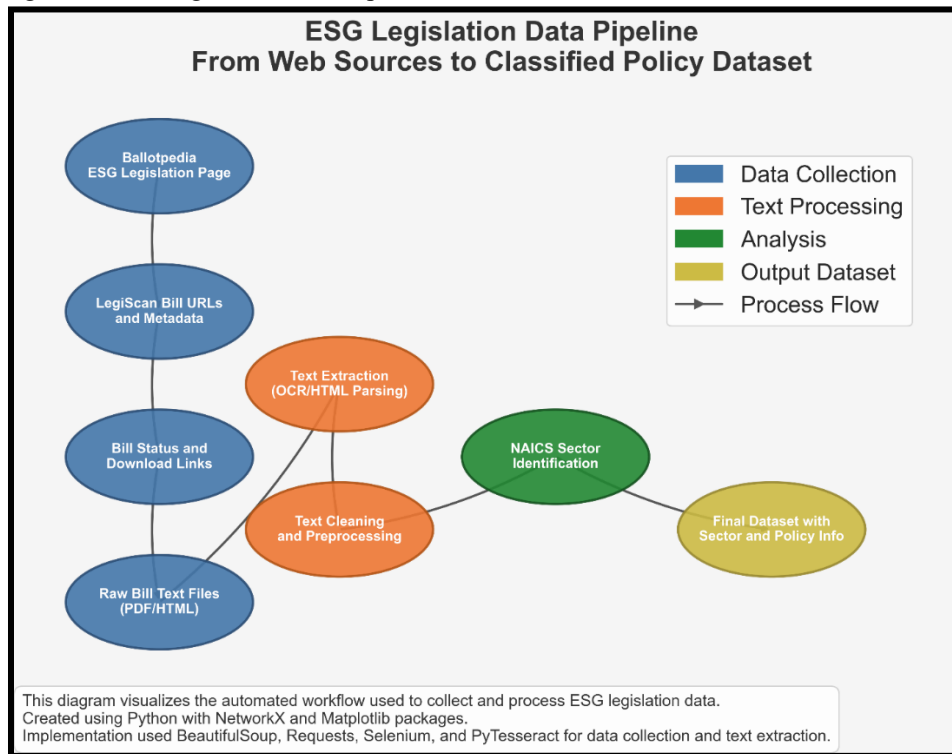
Legislative Data

The data collection for this research established a comprehensive pipeline to gather, process, and analyze ESG legislation across the United States (see figure 3). The process began with the Ballotpedia ESG Legislation Page, which served as the initial entry point for identifying relevant bills. This resource maintains a dedicated tracker for ESG-related legislation that provided valuable metadata including bill status (pro or anti-ESG), state of origin, and hyperlinks to full bill texts. A complementary source, LegiScan, offered official bill URLs, detailed metadata, and downloadable bill text files in various formats.

To extract data from these sources, I developed a web scraping architecture using the Python Requests library to access the Ballotpedia ESG tracker page, coupled with BeautifulSoup for HTML parsing. This approach allowed for systematic extraction of key bill information including the pro/anti ESG status, state jurisdiction, and bill hyperlinks to LegiScan. Upon accessing each LegiScan link, I employed BeautifulSoup to parse the HTML structure and extract additional metadata such as bill identification numbers, enactment dates, and current

status information. To acquire the full text of each bill, I implemented Selenium WebDriver to navigate LegiScan's interface and programmatically download the documents in their available formats, primarily PDF and HTML.

Figure 3. ESG Legislation Data Pipeline



Note: This automated approach enabled systematic collection and standardization of legislative texts, thus making the procedure replicable for future research.

The raw bill texts required significant processing before analysis. For PDF documents, I employed PyTesseract to convert image-based text into machine-readable format, a necessary step as many legislative documents were stored as scanned images rather than text-based PDFs. Following OCR conversion, I developed a text preprocessing module that removed extraneous numbers and whitespace, standardized formatting inconsistencies, eliminated boilerplate legislative language, and normalized text case and punctuation to ensure consistency across all documents.

To classify bills by their industrial relevance, I created a keyword-to-sector mapping dictionary that associated industry-specific terminology with NAICS (North American Industry Classification System) sector codes. This approach allowed for automated categorization of legislation based on its content. For example, terms like "electricity" and "generation" were mapped to sector 22 (Utilities), while "sustainable farming" was associated with sector 11 (Agriculture). Similarly, references to "mining," "waste management," and "remediation" were linked to sector 21 (Mining), and phrases like "educational institutions" and "curriculum" were connected to sector 61 (Educational Services).

The processed data was structured into a comprehensive dataset containing bill identifiers and URLs, state of origin, pro/anti ESG classification, enactment dates and current status, full bill text in cleaned format, NAICS sector codes (with multiple sectors where applicable), and ESG policy type classification. This structured dataset facilitated both quantitative and qualitative analyses, enabling examination of sectoral patterns in ESG legislation across different states and policy orientations. The multi-stage data pipeline ensured that the final dataset maintained high fidelity to the original legislative documents while providing standardized, analyzable content for subsequent research phases.

Data Summary

The dependent variables measuring sustainability performance demonstrate considerable variation across firms (see Table 4). The overall ESG index score has a mean of 32.13 (SD = 13.42), ranging from 6 to 91, indicating substantial heterogeneity in corporate sustainability practices. The environmental dimension exhibits the highest mean index score (28.45, SD = 17.84) and the widest range (0 to 99), suggesting greater variability in environmental performance

compared to other ESG dimensions. Social performance index scores average 26.98 (SD = 14.13), with a range from 0 to 92.

With respect to the primary independent variables of interest, 4.9% of firm-year observations are subject to anti-ESG state legislation, while 10.4% operate under pro-ESG regulatory frameworks. The mean board gender diversity index score is 51.02 (SD = 30.75), suggesting substantial variation in female representation on corporate boards, with scores ranging from 0 to 100. Board tenure index scores average 78.28 (SD = 28.44), also showing considerable variation across the sample.

The firm-level control variables indicate that my sample consists predominantly of established companies, with a mean age of 61.23 years (SD = 48.11), though the sample includes both newly established firms (minimum age of 1 year) and long-standing corporations (maximum age of 251 years). Firm size, measured as the natural logarithm of total assets, has a mean of 8.39 (SD = 1.77), with values ranging from 3.13 to 15.20, reflecting a diverse sample in terms of organizational scale.

Financial characteristics show that sample firms maintain an average leverage ratio of 1.42 (SD = 2.40), indicating that, on average, firms carry debt exceeding their shareholders' equity. Return on assets averages 1.5% (SD = 13.2%), with substantial variation ranging from -99.8% to 39.3%, suggesting the inclusion of both financially distressed and highly profitable companies. The mean Tobin's Q of 2.19 (SD = 2.03) indicates that, on average, the market values of these firms exceed their book values, though this metric also exhibits considerable variation (ranging from 0.49 to 27.90).

The wide ranges and substantial standard deviations across all variables underscore the heterogeneity of the sample, which enhances the generalizability of my findings across diverse

organizational contexts. This sample composition allows for robust examination of the relationships between state-level ESG legislation and corporate sustainability practices while controlling for firm-specific characteristics that prior literature has established as significant determinants of ESG performance.

Empirical Method

The primary independent variables consist of two binary indicators representing divergent policy approaches: AntiESGPolicy, which equals one if a firm is headquartered in a state that has enacted anti-legislation legislation, and zero otherwise; and ProESGPolicy, which equals one if a firm is headquartered in a state with legislation promoting or mandating ESG considerations, and zero otherwise. Board gender diversity, measured as the weighted percentage of female directors on the corporate board, serves as a theoretically significant moderating variable that may influence how firms respond to policy pressures.

Sample Selection

The empirical model specifically analyzes firms that were directly implicated in ESG legislation for a given state. For example, if Wyoming's ESG legislation does not implicate mining or oil industries, then mining and oil firms headquartered in Wyoming are excluded from the analysis sample. Firm observations are included in the treatment group only when two conditions are met: (1) the observation year is equal to or greater than the legislation's year of enactment, and (2) the legislation explicitly implicates the industry in which the firm operates. This refined approach ensures that the estimated policy effects capture the impact on precisely those firms that fall under the regulatory scope of state ESG policies, rather than assuming uniform effects across all firms within a state.

Variable Construction

The primary independent variables consist of two binary indicators representing divergent policy approaches: *AntiESGPolicy*, which equals one if a firm is headquartered in a state that has enacted legislation restricting ESG considerations in investments or corporate governance, and zero otherwise; and *ProESGPolicy*, which equals one if a firm is headquartered in a state with legislation promoting or mandating ESG considerations, and zero otherwise. Board gender diversity, measured as the weighted percentage of female directors on the corporate board, serves as a moderating variable that may influence how firms respond to policy pressures.

To isolate the effects of ESG policies and control for potential confounding factors, the model incorporates several firm-level controls: leverage (total debt divided by shareholders' equity), company size (natural logarithm of total assets), return on assets (net income scaled by total assets), Tobin's Q (market value of assets divided by book value of assets), company age (years since founding), board weighted average tenure index, and board weighted diversity index.

Model Specification

The baseline ordinary least squares regression model is specified as:

$$Y_{it} = \beta_0 + \beta_1 \text{AntiESGPolicy}_{st} + \beta_2 \text{ProESGPolicy}_{st} + \beta_k X_{it} + \alpha_i + \alpha_t + \varepsilon_{it} \quad (4)$$

where Y_{it} represents the ESG performance metric (overall, environmental, or social score) for firm i at time t . The coefficients β_1 and β_2 capture the direct effects of anti-ESG and pro-ESG policies, respectively. Vector X_{it} encompasses the firm-level control variables, while α_i and α_t represent year and company fixed effects, respectively, and ε_{it} is the error term.

To investigate the moderating role of board gender diversity on policy effects, I extend the model to include interaction terms:

$$\begin{aligned}
Y_{it} = & \beta_0 + \beta_1 \text{AntiESGPolicy}_{st} + \beta_2 \text{ProESGPolicy}_{st} \\
& + \beta_3 (\text{AntiESGPolicy} \times \text{BoardGenderDiversity})_{it} \\
& + \beta_4 (\text{ProESGPolicy} \times \text{BoardGenderDiversity})_{it} + \beta_k X_{it} + \alpha_i + \alpha_t + \varepsilon_{it}
\end{aligned} \tag{5}$$

In this expanded specification, β_3 and β_4 represent the differential effects of state ESG policies conditional on varying levels of board gender diversity. This approach enables the assessment of whether gender diversity moderates the effects of anti-ESG policies pro-ESG legislation. To address potential heteroscedasticity, I estimated the model using robust standard errors. This approach provides more conservative inference by accounting for potential violations of homoscedasticity assumptions. The inclusion of fixed effects helps mitigate concerns about omitted variable bias from unobserved year and company characteristics that might correlate with both ESG policies and firm performance.

Discussion

The empirical analysis reveals nuanced relationships between state-level ESG policies and corporate sustainability outcomes. These findings contribute to understanding the complex interplay between regulatory frameworks and firm ESG performance, while highlighting the moderating role of board composition.

Differential Policy Effects on ESG Dimensions

The results demonstrate that anti-ESG and pro-ESG policies have asymmetric effects on corporate sustainability performance, varying significantly across ESG dimensions (see Table 5). anti-ESG legislation shows a statistically significant negative impact on environmental performance ($\beta = -2.097$, $p < 0.01$) in the baseline model, with an even stronger effect when accounting for interaction terms ($\beta = -4.880$, $p < 0.01$). This substantial coefficient suggests that

anti-ESG policies may effectively discourage corporate environmental initiatives, potentially by reducing regulatory pressure and signaling state-level de-prioritization of environmental concerns.

In contrast, pro-ESG policies exhibit a positive and significant effect on overall ESG scores ($\beta = 0.708$, $p < 0.05$) and social performance ($\beta = 1.509$, $p < 0.01$) in the baseline model. This finding aligns with signaling theory (Connelly et al., 2011), suggesting that supportive regulatory environments encourage firms to enhance their ESG disclosures and performance to signal compliance with evolving norms. Surprisingly, pro-ESG policies show no significant effect on environmental performance, indicating these policies may be more effective at improving social governance aspects than addressing environmental challenges in the short term. Future research should incorporate a lag variable to account for strategic compliance that firms exhibit in the face of new regulation. As witnessed by Arvidsson and Dumay's (2022), firms initially enter into a minimal compliance strategy until enforcement mechanisms are better understood, at which point ESG performance begins to increase.

Table 5. Baseline Results

<i>Variables</i>	<i>Overall ESG Score</i>	<i>Environmental Score</i>	<i>Social Score</i>
<i>Anti ESG Policy (I=Yes)</i>	-0.0852 (0.411)	-2.097*** (0.635)	0.484 (0.514)
<i>Pro ESG Policy (I=Yes)</i>	0.708** (0.358)	-0.0948 (0.521)	1.509*** (0.411)
<i>Board Gender Diversity</i>	0.0454*** (0.00404)	0.0361*** (0.00569)	0.0331*** (0.00491)
<i>Board Average Tenure</i>	0.0206*** (0.00492)	0.00871 (0.00636)	0.0208*** (0.00583)
<i>Company Age</i>	-0.0612 (0.0661)	0.182** (0.0892)	0.373*** (0.0774)
<i>Company Size</i>	2.240*** (0.313)	2.760*** (0.446)	2.066*** (0.362)
<i>Leverage</i>	0.0583 (0.0572)	0.0998 (0.0811)	0.0494 (0.0646)
<i>Return on Assets</i>	2.117*** (0.752)	0.837 (1.116)	3.245*** (0.942)
<i>Tobin's Q</i>	0.211*** (0.0584)	0.376*** (0.103)	0.118* (0.0714)
<i>Constant</i>	12.51*** (3.720)	-9.254* (5.356)	-17.04*** (4.349)
<i>Company FE</i>	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes
<i>R-squared</i>	0.175	0.214	0.111

*Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. 2,938 unique companies, $n = 12,938$.*

Environmental vs. Social Impact: Resource Economy Considerations

A particularly notable finding is the stronger negative effect of anti-ESG policies on environmental performance compared to social metrics. This disparity supports congressman Jordan et al.'s (2023) assertion that anti-ESG legislation is economic for resource-extractive state economies. The concentration of negative effects in environmental dimensions rather than social aspects suggests these policies may specifically target environmental restrictions that could constrain resource extraction industries.

This asymmetry challenges Naeem et al.'s (2022) findings that environmentally sensitive firms benefit financially from stronger ESG performance. The significant negative coefficient for environmental scores under anti-ESG regimes indicates these policies may create environments where short-term economic considerations outweigh the long-term risk mitigation benefits identified by Naeem et al. (2022) and Alvarez-Perez and Fuentes (2024). By focusing primarily on immediate economic concerns over long-term sustainability, anti-ESG legislation appears to create conditions where firms reduce environmental commitments without necessarily altering social initiatives.

The findings expand upon Garrett and Ivanov's (2022) research on the financial implications of anti-ESG legislation in Texas. While their study documented increased borrowing costs following market exit by major financial institutions, this broader analysis reveals the direct impact on corporate sustainability performance. The pronounced negative effects on environmental scores align with their observed market disruptions, suggesting Anti-ESG policies create a dual burden: potentially increasing costs for citizens while simultaneously undermining environmental stewardship. This connection between financial consequences and diminished environmental performance further solidifies the economic-environmental trade-off that characterizes Anti-ESG policy environments.

Critical Mass Theory and ESG Policy Effectiveness

The interaction model reveals significant moderating effects of board gender diversity on the relationship between anti-ESG policies and sustainability performance (see Table 6). Specifically, the interaction term between anti-ESG policies and board gender diversity shows positive and significant coefficients across all ESG dimensions: overall ESG ($\beta = 0.0180$, $p < 0.05$), environmental ($\beta = 0.0521$, $p < 0.01$), and social ($\beta = 0.0293$, $p < 0.05$). These results

strongly support Critical Mass Theory (Kanter, 1977; Konrad et al., 2008), indicating that gender-diverse boards effectively buffer against negative regulatory pressures.

These findings extend Critical Mass Theory beyond its traditional application to boardroom dynamics by demonstrating its relevance to policy resilience. Boards with higher gender diversity appear more resistant to external pressures to reduce ESG commitments, particularly in environments characterized by Anti-ESG legislation. This aligns with Yarram and Adapa's (2021) observation that gender-diverse boards maintain stronger sustainability commitments even in challenging contexts. Interestingly, the interaction between pro-ESG policies and board gender diversity does not yield statistically significant results. This asymmetry suggests that while gender-diverse boards are particularly effective at resisting negative regulatory pressure, they may not necessarily amplify positive policy effects.

Table 6. Interaction Results

<i>Variables</i>	<i>Overall ESG Score</i>	<i>Environmental Score</i>	<i>Social Score</i>
<i>Anti ESG Policy (I=Yes)</i>	-1.042*	-4.880***	-1.076
	(0.560)	(0.844)	(0.756)
<i>Anti ESG Policy x Board Gender Diversity</i>	0.0180**	0.0521***	0.0293**
	(0.00909)	(0.0141)	(0.0121)
<i>Pro ESG Policy (I=Yes)</i>	0.248	-0.580	0.863
	(0.645)	(1.020)	(0.743)
<i>Pro ESG Policy x Board Gender Diversity</i>	0.00693	0.00749	0.00976
	(0.00948)	(0.0150)	(0.0107)
<i>Board Gender Diversity</i>	0.0440***	0.0332***	0.0310***
	(0.00415)	(0.00571)	(0.00508)
<i>Board Average Tenure</i>	0.0205***	0.00859	0.0207***
	(0.00492)	(0.00635)	(0.00583)
<i>Company Age</i>	-0.0571	0.191**	0.379***
	(0.0662)	(0.0892)	(0.0775)
<i>Company Size</i>	2.251***	2.781***	2.082***
	(0.313)	(0.446)	(0.363)
<i>Leverage</i>	0.0585	0.100	0.0496
	(0.0573)	(0.0812)	(0.0646)
<i>Return on Assets</i>	2.147***	0.913	3.293***
	(0.751)	(1.114)	(0.940)
<i>Tobin's Q</i>	0.212***	0.377***	0.120*
	(0.0585)	(0.103)	(0.0716)
<i>Company FE</i>	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes
<i>R-squared</i>	0.19	0.21	0.111

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. 2,938 unique companies. $n = 12,938$.

Firm Characteristics as Determinants of ESG Performance

Consistent with existing literature, several firm characteristics emerge as significant determinants of ESG performance across both models. Company size demonstrates the strongest positive association with all ESG dimensions (β ranging from 2.066 to 2.781, $p < 0.01$), confirming findings by Khaled et al. (2021) and Lestari and Adhariani (2022) that larger firms typically invest more resources in sustainability initiatives. This substantial effect underscores

the resource advantages that larger corporations maintain in pursuing comprehensive ESG programs.

Board gender diversity consistently shows significant positive associations with all ESG dimensions in both models (β ranging from 0.0310 to 0.0454, $p < 0.01$), aligning with previous research by Chang et al. (2024) and Shakil et al. (2021). Similarly, board tenure demonstrates positive associations with overall ESG and social scores ($\beta \approx 0.020$, $p < 0.01$), supporting the expertise hypothesis proposed by Schnake et al. (2011) and validated by Andreassen and Bukhari (2024).

Financial performance indicators show varied relationships with ESG outcomes. Return on Assets positively associates with overall ESG and social dimensions ($\beta = 2.117$ and 3.245 respectively, $p < 0.01$), confirming Garcia and Orsato's (2020) finding that profitability enables greater sustainability investments. Tobin's Q demonstrates positive relationships across all ESG dimensions, supporting Ortas et al.'s (2015) assertion that market valuation reflects and reinforces ESG commitments. Interestingly, leverage shows no significant relationship with ESG performance, diverging somewhat from Khaled et al.'s (2024) findings in emerging markets.

Company age presents an interesting pattern, showing insignificant relationships with overall ESG scores but positive associations with environmental ($\beta = 0.182$, $p < 0.05$) and social dimensions ($\beta = 0.373$, $p < 0.01$). This suggests that while mature firms may have established environmental and social programs, they do not necessarily outperform younger firms in overall ESG integration, perhaps due to governance structures that may be less responsive to emerging ESG concerns.

CHAPTER V

CONCLUSION

This study examined the relationship between state-level ESG legislation and corporate ESG performance in the United States, revealing important implications for policy design, corporate governance, and sustainable development. The findings underscore how polarized regulatory landscapes shape corporate ESG behaviors, while highlighting the crucial role of board composition in mediating these effects. They also provide a pertinent foundation for future research on U.S. ESG legislation.

A particularly noteworthy finding is that anti-ESG legislation exerts a substantially negative impact on firms' environmental performance. This disruptive behavior is a parallel the findings by Garrett and Ivanov's (2022). Although this outcome raises concerns, further research is needed to isolate the specific mechanisms through which such legislation signals firms to reduce their environmental performance. At present, there is clear evidence that anti-ESG legislation increases the stock returns of firms with poor environmental track records, suggesting that these measures may incentivize weaker environmental outcomes (Tang et al., 2024). Given this study's broad approach to ESG policy, it is not yet possible to pinpoint which state-level anti-ESG measures convey the strongest signals. Future research should therefore prioritize more focused legislative analyses.

This study makes several contributions to theory and practice. First, it extends Critical Mass Theory beyond boardroom dynamics to policy resilience, showing how gender-diverse boards buffer against negative regulatory pressures. The interaction terms between anti-ESG policies and board gender diversity show consistently positive and significant effects across all ESG dimensions: overall ESG ($\beta = 0.0180$, $p < 0.05$), environmental ($\beta = 0.0521$, $p < 0.01$), and

social ($\beta = 0.0293$, $p < 0.05$). This extends prior work by Kanter (1977) and Konrad et al. (2008) by demonstrating that diverse governance structures not only improve internal decision-making but also enhance resilience against external policy pressures. Due to the weighting of the S&P global data, this study was not able to describe the true gender and tenure traits of corporate boards. It's in my suggestion, that future research understand the limitations of S&P Global's CSA data to derive absolute measures of corporate social structures for analysis. None the less, the moderating effects gender warrant future analysis.

Such analyses can be resource-intensive, requiring specialized expertise and considerable time to assess, hypothesize, and test how legislative language influences real-world practices. Large language models (LLMs) offer a promising tool for overcoming these challenges by rapidly processing and synthesizing complex legal texts. Their utility has already been demonstrated in the domain of ESG analysis, where they have been used to evaluate extensive, language-dense corporate disclosures (Zou et al., 2025). Leveraging LLMs in future investigations could significantly advance our understanding of how specific legislative provisions shape ESG performance, thereby refining both academic inquiry and policy development in this critical area.

Future methods should investigate an instrumental variables approach. The negative coefficient on anti-ESG policy ($\beta = -2.097$) for environmental performance may reflect omitted variable bias if unobserved factors influence both policy adoption and ESG outcomes. Future studies could employ instrumental variables that predict policy adoption but are unrelated to firm-level ESG performance, such as political composition changes unrelated to ESG issues or historical state regulatory patterns.

Lastly, for policymakers, these findings suggest that anti-ESG legislation may achieve political objectives while generating environmental consequences. The substantial negative coefficients for environmental performance ($\beta = -2.097$ to -4.880) indicate potential long-term costs that should be weighed against immediate economic benefits. Additionally, the minimal impact on social performance suggests opportunities for more balanced policy approaches that address economic concerns without compromising all dimensions of corporate responsibility.

CHAPTER VII

REFERENCES

- Abideen, Z. U. (2023). Firm performance in China: an analysis of the interplay between financial leverage, firm liquidity, and firm size. *Journal of Accounting and Finance in Emerging Economies*, 9(3), 141–150. <https://doi.org/10.26710/jafee.v9i3.2649>
- Achabou, M. A. (2020). The effect of perceived CSR effort on consumer brand preference in the clothing and footwear sector. *European Business Review*, 32(2), 317–347. <https://doi.org/10.1108/EBR-11-2018-0198>
- Adams, R. B., & Ferreira, D. (2009). Women in the boardroom and their impact on governance and performance. *Journal of Financial Economics*, 94(2), 291–309. <https://doi.org/10.1016/j.jfineco.2008.10.007>
- An Act to Amend the Law Concerning Environmental, Social Justice, or Governance Scores, H.B. 1845, 94th General Assembly, Regular Session (2023). <https://legiscan.com/ar/text/hb1845/2023>
- An Act to Amend Title 12 Of The Delaware Code Relating To Decedents Estates And Fiduciary Relations, H.B. 334, 150th General Assembly, Regular Session (2020). <https://legiscan.com/de/text/hb334/2019>
- An Act Regarding the Procurement of Energy from Offshore Wind Resources, Me. Pub. L. No. 2023, ch. 481 (2023). <https://legiscan.com/me/text/ld1895/2023>
- An Act to Require the State To Divest Itself of Assets Invested in the Fossil Fuel Industry, Me. Pub. L. No. 2021, ch. 231 (2021). <https://legiscan.com/me/text/ld99/2021>

- Ahmad, N., Mobarek, A., & Roni, N. N. (2021). Revisiting the impact of ESG on financial performance of FTSE350 UK firms: static and dynamic panel data analysis. *Cogent Business & Management*, 8(1), 1900500. <https://doi.org/10.1080/23311975.2021.1900500>
- Alexander, J. (2007). Environmental sustainability versus profit maximization: overcoming systemic constraints on implementing normatively preferable alternatives. *Journal of Business Ethics*, 76(2), 155–162. <https://doi.org/10.1007/s10551-006-9264-5>
- AlJanadi, Y. (2025). Gender diversity and disclosure: a meta-analysis. *International Journal of Disclosure and Governance*. <https://doi.org/10.1057/s41310-024-00285-w>
- Aluchna, M., Roszkowska-Menkes, M., & Kamiński, B. (2023). From talk to action: the effects of the non-financial reporting directive on esg performance. *Meditari Accountancy Research*, 31(7), 1–25. <https://doi.org/10.1108/MEDAR-12-2021-1530>
- Alvarez-Perez, H., & Fuentes, R. (2024). ESG disclosure and financial performance in debt market: evidence from the oil and gas industry. *Academia Revista Latinoamericana de Administración*, 37(4), 634–653. <https://doi.org/10.1108/ARLA-07-2024-0135>
- Amel-Zadeh, A., & Serafeim, G. (2018). Why and how investors use ESG information: evidence from a global survey. *Financial Analysts Journal*, 74(3), 87–103. <https://doi.org/10.2469/faj.v74.n3.2>
- Andreassen, S. D., & Bukhari, S. K. U. H. (2024). Impacts of board characteristics on ESG performance in the industrial sector: a quantitative study with evidence from Scandinavia. Oslo Metropolitan University. <https://hdl.handle.net/11250/3167038>
- Annunziata, F., & Siri, M. (2025). The EU regulation on ESG ratings: fit for purpose? *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.5138977>

- Anti-ESG Act, S.B. 261, 2023 Regular Session (2023). <https://legiscan.com/al/text/sb261/2023>
- Arif, M., Gan, C., & Nadeem, M. (2022). Regulating non-financial reporting: evidence from European firms' environmental, social and governance disclosures and earnings risk. *Meditari Accountancy Research*, 30(3), 495–523. <https://doi.org/10.1108/MEDAR-11-2020-1086>
- Arun, T., Girardone, C., & Piserà, S. (2022). ESG issues in emerging markets and the role of banks. In D. Nguyen (Ed.), *Handbook of Banking and Finance in Emerging Markets* (pp. 321–344). Edward Elgar Publishing. <https://doi.org/10.4337/9781800880900.00025>
- Arvidsson, S., & Dumay, J. (2022). Corporate ESG reporting quantity, quality and performance: where to now for environmental policy and practice? *Business Strategy and the Environment*, 31(3), 1091–1110. <https://doi.org/10.1002/bse.2937>
- Assembly Bill 979, Cal. Corp. Code §§ 301.3, 301.4, 2115.6 (2020). <https://legiscan.com/ca/text/ab979/id/2211047>
- Avetisyan, E., & Hockerts, K. (2017). The consolidation of the ESG rating industry as an enactment of institutional retrogression. *Business Strategy and the Environment*, 26(3), 316–330. <https://doi.org/10.1002/bse.1919>
- Ballinger, J. (1992). The new free-trade heel: Nike's profits jump on the backs of Asian workers. *Harper's*, 285(1707), 46. Periodicals Archive Online.
- Bartley, T., & Child, C. (2011). Movements, markets and fields: the effects of anti-sweatshop campaigns on U.S. Firms, 1993-2000. *Social Forces*, 90(2), 425–451. <https://doi.org/10.1093/sf/sor010>
- Beasley, M. S. (1996). An empirical analysis of the relation between the board of director composition and financial statement fraud. *The Accounting Review*, 71(4), 443–465.

- Berg, F., Kölbel, J. F., & Rigobon, R. (2022). Aggregate confusion: the divergence of ESG ratings. *Review of Finance*, 26(6), 1315–1344. <https://doi.org/10.1093/rof/rfac033>
- Cambrea, D. R., Paolone, F., & Cucari, N. (2023). Advisory or monitoring role in ESG scenario: which women directors are more influential in the Italian context? *Business Strategy and the Environment*, 32(7), 4299–4314. <https://doi.org/10.1002/bse.3366>
- Canavan, J., Jones, B., & Potter, M. J. (2004). Board tenure: how long is too long? *Directors & Boards*, 28(2), 39–42.
- Celli, M., Arduini, S., & Beck, T. (2024). Corporate sustainability reporting directive and his future application scenario for Italian SMES. *International journal of business and management*, 19(4), 44. <https://doi.org/10.5539/ijbm.v19n4p44>
- Chan, C.-Y., Chou, D.-W., & Lo, H.-C. (2017). Do financial constraints matter when firms engage in CSR? *The North American Journal of Economics and Finance*, 39, 241–259. <https://doi.org/10.1016/j.najef.2016.10.009>
- Chang, Y., Wu, K.-T., Lin, S.-H., & Lin, C.-J. (2024). Board gender diversity and corporate social responsibility. *International Journal of Corporate Social Responsibility*, 9(1), 7. <https://doi.org/10.1186/s40991-024-00095-x>
- Chatterji, A. K., Durand, R., Levine, D. I., & Touboul, S. (2016). Do ratings of firms converge? Implications for managers, investors and strategy researchers. *Strategic Management Journal*, 37(8), 1597–1614. <https://doi.org/10.1002/smj.2407>
- Chen, Z., Lai, Y., & Zhou, Y. (2023). Regional heterogeneity analysis of ESG and corporate value. *Advances in Economics, Management and Political Sciences*, 63(1), 231–238. <https://doi.org/10.54254/2754-1169/63/20231428>

- Cicchiello, A. F., Marrazza, F., & Perdichizzi, S. (2023). Non-financial disclosure regulation and environmental, social, and governance. *Corporate Social Responsibility and Environmental Management*, 30(3), 1121–1128. <https://doi.org/10.1002/csr.2408>
- Climate Corporate Data Accountability Act, No. Senate Bill No. 253, California State Legislature (2023).
https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202320240SB253
- Connelly, B. L., Certo, S. T., Ireland, R. D., & Reutzel, C. R. (2011). Signaling theory: a review and assessment. *Journal of Management*, 37(1), 39–67.
<https://doi.org/10.1177/0149206310388419>
- Corporations LGBTQ Directors, Ill. Public Act 102-0223 (2021).
<https://legiscan.com/il/text/sb1730/2021>
- Cremona, B. M., & Passador, M. L. (2019). What about the future of european banks? Board characteristics and ESG impact. *Banking & Insurance eJournal*.
<https://api.semanticscholar.org/CorpusID:214427341>
- Čufar, M., & Belak, J. (2024). Beyond financials: understanding the implications of NFRD and CSRD on non-financial reporting. *Challenges in the Turbulent Economic Environment and Organizations' Sustainable Development*, 185–194.
<https://doi.org/10.18690/um.epf.5.2024.18>
- Cuomo, F., Gaia, S., Girardone, C., & Piserà, S. (2024). The effects of the EU non-financial reporting directive on corporate social responsibility. *The European Journal of Finance*, 30(7), 726–752. <https://doi.org/10.1080/1351847X.2022.2113812>

- De Giuli, M. E., Grechi, D., & Tanda, A. (2024). What do we know about ESG and risk? A systematic and bibliometric review. *Corporate Social Responsibility and Environmental Management*, 31(2), 1096–1108. <https://doi.org/10.1002/csr.2624>
- Dezsö, C. L., & Ross, D. G. (2012). Does female representation in top management improve firm performance? A panel data investigation. *Strategic Management Journal*, 33(9), 1072–1089. <https://doi.org/10.1002/smj.1955>
- Eccles, R. G., Lee, L.-E., & Strohle, J. C. (2020). The social origins of ESG: an analysis of Innovest and KLD. *Organization & Environment*, 33(4), 575–596. <https://doi.org/10.1177/1086026619888994>
- Enacted state ESG legislation by trifecta status, 2020-2024. (2024). Ballotpedia. https://ballotpedia.org/Enacted_state_ESG_legislation_by_trifecta_status,_2020-2024
- Energy Discrimination Elimination Act of 2022, H.B. 2034, 2022 Regular Session (2022). <https://legiscan.com/ok/text/hb2034/2022>
- Escrig-Olmedo, E., Fernández-Izquierdo, M., Ferrero-Ferrero, I., Rivera-Lirio, J., & Muñoz-Torres, M. (2019). Rating the raters: evaluating how ESG rating agencies integrate sustainability principles. *Sustainability*, 11(3), 915. <https://doi.org/10.3390/su11030915>
- European Parliament. (2022). Directive (EU) 2022/2464. <https://eur-lex.europa.eu/eli/dir/2022/2464/oj/eng>
- Executive Order Review Process Amendments, Utah Code Ann. § 63C-4a-204 (2021). <https://legiscan.com/ut/text/hb0415/2021>
- Fair Labor Association. (2003). First public report: towards improving workers' lives. https://www.fairlabor.org/wp-content/uploads/2022/01/2003_annual_public_report.pdf

- Ferrer, E., López-Arceiz, F. J., & Del Rio, C. (2020). Sustainability disclosure and financial analysts' accuracy: The European case. *Business Strategy and the Environment*, 29(8), 2939–2952. <https://doi.org/10.1002/bse.2549>
- Friede, G., Busch, T., & Bassen, A. (2015). ESG and financial performance: aggregated evidence from more than 2000 empirical studies. *Journal of Sustainable Finance & Investment*, 5(4), 210–233. <https://doi.org/10.1080/20430795.2015.1118917>
- Friedman, M. (1957). The social responsibility of business is to increase its profits. In W. C. Zimmerli, M. Holzinger, & K. Richter (Eds.), *Corporate Ethics and Corporate Governance* (pp. 173–178). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-540-70818-6_14
- Friske, W., Hoelscher, S. A., & Nikolov, A. N. (2023). The impact of voluntary sustainability reporting on firm value: insights from signaling theory. *Journal of the Academy of Marketing Science*, 51(2), 372–392. <https://doi.org/10.1007/s11747-022-00879-2>
- Garcia, A. S., & Orsato, R. J. (2020). Testing the institutional difference hypothesis: a study about environmental, social, governance, and financial performance. *Business Strategy and the Environment*, 29(8), 3261–3272. <https://doi.org/10.1002/bse.2570>
- Garrett, D., & Ivanov, I. (2022). Gas, guns, and governments: financial costs of anti-ESG policies. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4123366>
- Gholami, A., Murray, P. A., & Sands, J. (2022). Environmental, social, governance & financial performance disclosure for large firms: is this different for SME firms? *Sustainability*, 14(10), 6019. <https://doi.org/10.3390/su14106019>
- Government and Corporate Activism Act, H.B. 003, 2023 Regular Session (2023). <https://legiscan.com/fl/text/h0003/2023>

- Granovetter, M. (1978). Threshold models of collective behavior. *American Journal of Sociology*, 83(6), 1420–1443. <https://doi.org/10.1086/226707>
- Hao, N., Dragomir, V. D., & Radu, O. M. (2023). Effects on corporate stakeholders and limitations of the implementation of the Non-Financial Reporting Directive (2014/95/EU). *Journal of Accounting and Management Information Systems*, 22(4). <https://doi.org/10.24818/jamis.2023.04002>
- Hendricks, K. B., & Singhal, V. R. (2001). Firm characteristics, total quality management, and financial performance. *Journal of Operations Management*, 19(3), 269–285. [https://doi.org/10.1016/S0272-6963\(00\)00049-8](https://doi.org/10.1016/S0272-6963(00)00049-8)
- Hess, D. J., Mai, Q. D., & Brown, K. P. (2016). Red states, green laws: ideology and renewable energy legislation in the United States. *Energy Research & Social Science*, 11, 19–28. <https://doi.org/10.1016/j.erss.2015.08.007>
- House Bill 457, H.B. 457, 2023 Regular Session (2023). <https://legiscan.com/nh/text/hb457/2023>
- House Bill 1429, H.B. 1429, 68th Legislative Assembly (2023). <https://legiscan.com/nd/text/hb1429/2023>
- House Concurrent Resolution 110, H.C.R. 110, 2023 Regular Session (2023). <https://legiscan.com/la/text/hcr110/id/2812806>
- Huang, D. Z. X. (2021). Environmental, social and governance (ESG) activity and firm performance: A review and consolidation. *Accounting & Finance*, 61(1), 335–360. <https://doi.org/10.1111/acfi.12569>
- Huang, S., & Hilary, G. (2018). Zombie board: board tenure and firm performance. *Journal of Accounting Research*, 56(4), 1285–1329. <https://doi.org/10.1111/1475-679X.12209>

Idaho Code, H.B. 191, 67th Legislature, 1st Regular Session (2023).

<https://legiscan.com/id/text/h0191/2023>

Jonsdottir, B., Sigurjonsson, T. O., Johannsdottir, L., & Wendt, S. (2022). Barriers to using ESG data for investment decisions. *Sustainability*, 14(9), 5157.

<https://doi.org/10.3390/su14095157>

Jordan, J., Massie, T., & Bishop, D. (2023). Letter to Michael R. Bloomberg and Mary L. Schapiro regarding potential antitrust violations by GFANZ and NZAM. U.S. house committee on the judiciary. <https://judiciary.house.gov/sites/evo-subsites/republicans-judiciary.house.gov/files/evo-media-document/2023-07-06-jdj-tm-db-to-bloomberg-schapiro-gfanz.pdf>

Kanter, R. M. (1977). *Men and women of the corporation*. Basic Books.

Katmon, N., Mohamad, Z. Z., Norwani, N. M., & Farooque, O. A. (2019). Comprehensive board diversity and quality of corporate social responsibility disclosure: evidence from an emerging market. *Journal of Business Ethics*, 157(2), 447–481.

<https://doi.org/10.1007/s10551-017-3672-6>

Khaled, R., Ali, H., & Mohamed, E. K. A. (2021). The sustainable development goals and corporate sustainability performance: mapping, extent and determinants. *Journal of Cleaner Production*, 311, 127599. <https://doi.org/10.1016/j.jclepro.2021.127599>

Kim, S., & Li, Z. (Frank). (2021). Understanding the impact of ESG practices in corporate finance. *Sustainability*, 13(7), 3746. <https://doi.org/10.3390/su13073746>

Konrad, A. M., Kramer, V., & Erkut, S. (2008). Critical mass: the impact of three or more women on corporate boards. *Organizational Dynamics*, 37(2), 145–164.

<https://doi.org/10.1016/j.orgdyn.2008.02.005>

- Lee, J. H., Choi, D., Choi, H., & Han, S. H. (2024). Corporate bond market reaction to the mandatory ESG disclosure act: is sustainium sustainable?. *Asia-Pacific Journal of Financial Studies*, 53(5), 596–625. <https://doi.org/10.1111/ajfs.12484>
- Legislative Services Agency. (2023). Fiscal Impact Statement for HB 1008. Legislative Services Agency, Office of Fiscal and Management Analysis. <https://web.archive.org/web/20230214101631/https://d37sr56shkhro8.cloudfront.net/pdf-documents/123/2023/house/bills/HB1008/fiscal-notes/HB1008.02.COMH.FN001.pdf>
- Lestari, N. I. G., & Adhariani, D. (2022). Can intellectual capital contribute to financial and non-financial performances during normal and crisis situations? *Business Strategy & Development*, 5(4), 390–404. <https://doi.org/10.1002/bsd2.206>
- Lombardi, R., Cosentino, A., Sura, A., & Galeotti, M. (2022). The impact of the EU directive on non-financial information: novel features of the Italian case. *Meditari Accountancy Research*, 30(6), 1419–1448. <https://doi.org/10.1108/MEDAR-06-2019-0507>
- Matuszak, Ł., & Róžańska, E. (2021). Towards 2014/95/EU directive compliance: the case of Poland. *Sustainability Accounting, Management and Policy Journal*, 12(5), 1052–1076. <https://doi.org/10.1108/SAMPJ-02-2020-0042>
- Mbanyele, W., Huang, H., Li, Y., Muchenje, L. T., & Wang, F. (2022). Corporate social responsibility and green innovation: Evidence from mandatory CSR disclosure laws. *Economics Letters*, 212, 110322. <https://doi.org/10.1016/j.econlet.2022.110322>
- Memon, M. A. (2024). Control variables: a review and proposed guidelines. *Journal of Applied Structural Equation Modeling*, 8(2), 1–14. [https://doi.org/10.47263/JASEM.8\(2\)01](https://doi.org/10.47263/JASEM.8(2)01)

- Mervelskemper, L., & Streit, D. (2017). Enhancing market valuation of ESG performance: is integrated reporting keeping its promise? *Business Strategy and the Environment*, 26(4), 536–549. <https://doi.org/10.1002/bse.1935>
- Ministry of Ecology and Environment. (2021). Management measures for disclosure of enterprise environmental information in accordance with the law. https://www.mee.gov.cn/xxgk2018/xxgk/xxgk02/202112/t20211221_964837.html?mc_cid=45e6a7ad33
- Miroshnychenko, I., Barontini, R., & Testa, F. (2017). Green practices and financial performance: a global outlook. *Journal of Cleaner Production*, 147, 340–351. <https://doi.org/10.1016/j.jclepro.2017.01.058>
- Mittelstaedt, B. (2024). Measuring the effectiveness of carbon pricing: the impact on corporate ESG performance and carbon emissions. <https://digitalcommons.cwu.edu/etd/1941>
- Murphy, D., & Mathew, D. (2001). Nike and global labour practices. *New Academy of Business*. <http://insight.cumbria.ac.uk/id/eprint/5584/>
- Naeem, N., Cankaya, S., & Bildik, R. (2022). Does ESG performance affect the financial performance of environmentally sensitive industries? A comparison between emerging and developed markets. *Borsa Istanbul Review*, 22, S128–S140. <https://doi.org/10.1016/j.bir.2022.11.014>
- Niu, B. (2024). Government environmental protection expenditure and national ESG performance: Global evidence. *Innovation and Green Development*, 3(2), 100117. <https://doi.org/10.1016/j.igd.2023.100117>
- Olson, M. (1965). *The logic of collective action: public goods and the theory of groups*. Harvard University Press.

- Ortas, E., Álvarez, I., & Garayar, A. (2015). The environmental, social, governance, and financial performance effects on companies that adopt the United Nations Global Compact. *Sustainability*, 7(2), 1932–1956. <https://doi.org/10.3390/su7021932>
- Pearce, E. (1957). History of the standard industrial classification. Office of Statistical Standards.
- Peng, P., & Sun, M. (2024). Government subsidies and corporate environmental, social and governance performance: evidence from companies of China. *International Studies of Economics*, 19(3), 374–405. <https://doi.org/10.1002/ise3.79>
- Pension Investments, H.B. 1008, 123rd General Assembly, 1st Regular Session (2023). <https://legiscan.com/in/text/hb1008/2023>
- Phillips, R. L., & Ormsby, R. (2016). Industry classification schemes: an analysis and review. *Journal of Business & Finance Librarianship*, 21(1), 1–25. <https://doi.org/10.1080/08963568.2015.1110229>
- Post, C., & Byron, K. (2015). Women on boards and firm financial performance: a meta-analysis. *Academy of Management Journal*, 58(5), 1546–1571. <https://doi.org/10.5465/amj.2013.0319>
- Prohibiting Banks or Businesses From Using Social Credit Scores, H.B. 1469, 2022 Regular Session (2022). <https://legiscan.com/nh/text/hb1469/id/2462053>
- Prohibition on Public Entities From Contracting With Companies That Boycott Energy, Fossil Fuel, Firearms, and Ammunition Industries Act, S.B. 62, 94th General Assembly, Regular Session (2023). <https://legiscan.com/ar/text/sb62/2023>
- Public Act 102-0297, S.B. 0653, 102nd General Assembly (2021). <https://legiscan.com/il/text/sb0653/2021>

Public Contract Requirements, S.B. 97, 2023 General Session (2023).

<https://legiscan.com/ut/text/sb0097/2023>

Public Employee Retirement Systems: Prohibited Investments: Turkey, S.B. 1089, Chapter 541

(2022). <https://legiscan.com/ca/text/sb1089/id/2607909>

Public Retirement Systems Investment Authority Law, H.B. 481, 2024 Regular Session (2024).

<https://legiscan.com/ga/text/hb481/2023>

Radu, O. M., Dragomir, V. D., & Hao, N. (2023). Company-level factors of non-financial

reporting quality under a mandatory regime: a systematic review of empirical evidence in

the European union. *Sustainability*, 15(23), 16265. <https://doi.org/10.3390/su152316265>

Rhode Island Secure Choice Retirement Savings Program Act, R.I. Gen. Laws ch. 35-23 (2024).

<https://legiscan.com/ri/text/s2045/2024>

Romano, R. (1993). Public pension fund activism in corporate governance reconsidered.

Columbia Law Review, 93, 795–827.

Saxena, A., Singh, R., Gehlot, A., Akram, S. V., Twala, B., Singh, A., Montero, E. C., &

Priyadarshi, N. (2022). Technologies empowered environmental, social, and governance

(ESG): An Industry 4.0 Landscape. *Sustainability*, 15(1), 309.

<https://doi.org/10.3390/su15010309>

Schanberg, S. H. (1996, June). Six cents an hour. *Life Magazine*, Volcano – It could happen

sooner than you think.

Senate Bill 16, S.B. 23-016, 74th General Assembly, Regular Session (2023).

<https://legiscan.com/co/text/sb016/2023>

Senate Bill 261, 2023-2024 Regular Session (2023).

<https://legiscan.com/ca/text/sb261/id/2670346>

- Senate Bill 833, 88th Legislature, Relating to consideration by insurers of certain prohibited criteria for ratemaking (2023). <https://legiscan.com/tx/text/sb833/2023>
- Senate Bill No. 655, Chapter 390, Cal. Ins. Code §§ 926-927.3 (2021).
<https://legiscan.com/ca/text/sb655/id/2434862>
- Shakil, M. H. (2021). Environmental, social and governance performance and financial risk: moderating role of ESG controversies and board gender diversity. *Resources Policy*, 72, 102144. <https://doi.org/10.1016/j.resourpol.2021.102144>
- Sharma, V. D., & Iselin, E. R. (2012). The association between audit committee multiple-directorships, tenure, and financial misstatements. *AUDITING: A Journal of Practice & Theory*, 31(3), 149–175. <https://doi.org/10.2308/ajpt-10290>
- Shen, H., Lin, H., Han, W., & Wu, H. (2023). ESG in China: A review of practice and research, and future research avenues. *China Journal of Accounting Research*, 16(4), 100325. <https://doi.org/10.1016/j.cjar.2023.100325>
- Smith, A. (1776). Chapter I.: Of the Principle of the Commercial or Mercantile System. In *Inquiry into the Nature & Causes of the Wealth of Nations* (pp. 234–247). Project Gutenberg Literary Archive Foundation.
- Solakoglu, M. N., & Demir, N. (2016). The role of firm characteristics on the relationship between gender diversity and firm performance. *Management Decision*, 54(6), 1407–1419. <https://doi.org/10.1108/MD-02-2015-0075>
- S&P Global. (2024a). CSA 2024 sample questionnaire. S&P Global.
https://web.archive.org/web/20250306193812/https://portal.s1.spglobal.com/survey/documents/CSA_2024_Sample_Questionnaire_BNK.pdf

- S&P Global. (2024b). S&P Global ESG Scores Methodology. S&P Global.
https://web.archive.org/web/20250306193904/https://portal.s1.spglobal.com/survey/documents/spglobal_esg_scores_methodology.pdf
- Spence, M. (1973). Job market signaling. *The Quarterly Journal of Economics*, 87(3), 355.
<https://doi.org/10.2307/1882010>
- Spence, M. (2002). Signaling in retrospect and the informational structure of markets. *American Economic Review*, 92(3), 434–459. <https://doi.org/10.1257/00028280260136200>
- State Government Employee Retirement Protection Act, H.B. 1253, 94th General Assembly, Regular Session (2023). <https://legiscan.com/ar/text/hb1253/2023>
- State Retirement and Pension System - Investment Climate Risk - Fiduciary Duties, Md. Code Ann., State Pers. & Pens. § 21-116.1 (2022). <https://legiscan.com/md/text/hb740/2022>
- Tang, O., Shi, X., & Jiu, L. (2024). Value creation or political trick? An event study on anti-ESG regulations. *Finance Research Letters*, 65, 105530.
<https://doi.org/10.1016/j.frl.2024.105530>
- Tennessee Investment Act, Tenn. Pub. Ch. No. 471 (2023).
<https://legiscan.com/tn/text/hb1286/id/2817571>
- Texas SB 13 (2021). <https://legiscan.com/TX/bill/SB13/2021>
- UN Environment Programme – Finance Initiative. (2004). Who cares wins – the global compact connecting financial markets to a changing world.
https://www.unepfi.org/fileadmin/events/2004/stocks/who_cares_wins_global_compact_2004.pdf

- Venturelli, A., Fasan, M., & Pizzi, S. (2022). Guest editorial Rethinking non-financial reporting in Europe: challenges and opportunities in revising Directive 2014/95/EU. *Journal of Applied Accounting Research*, 23(1), 1–7. <https://doi.org/10.1108/JAAR-02-2022-265>
- Wald, J. K. (1999). How firm characteristics affect capital structure: an international comparison. *Journal of Financial Research*, 22(2), 161–187. <https://doi.org/10.1111/j.1475-6803.1999.tb00721.x>
- Wang, K. T., & Li, D. (2016). Market reactions to the first-time disclosure of corporate social responsibility reports: evidence from china. *Journal of Business Ethics*, 138(4), 661–682. <https://doi.org/10.1007/s10551-015-2775-1>
- Wang, L. (2024). Challenges and opportunities of ESG integration in financial operations. *Modern Management Science & Engineering*, 6(1), p162. <https://doi.org/10.22158/mmse.v6n1p162>
- West Virginia Investment Management Board and Board of Treasury Investments Shareholder Voting Requirements Act, W. Va. Code §§ 12-6-11a, 12-6C-13 (2023). <https://legiscan.com/wv/text/hb2862/2023>
- Williams, O. F. (2007). The UN Global Compact: the challenge and the promise. In W. C. Zimmerli, M. Holzinger, & K. Richter (Eds.), *Corporate ethics and corporate governance* (pp. 287–308). Springer. https://doi.org/10.1007/978-3-540-70818-6_20
- World Commission on Environment and Development. (1987). *Our common future: report of the world commission on environment and development*.
- Wu, L., Liu, S., Qi, L., & Lin, D. (2025). Mandatory disclosure and corporate ESG performance: evidence from China’s “explanation for nondisclosure” requirement. *Corporate Social*

- Responsibility and Environmental Management, 32(1), 176–191.
<https://doi.org/10.1002/csr.2944>
- Wu, S., Li, X., Du, X., & Li, Z. (2022). The impact of ESG performance on firm value: the moderating role of ownership structure. *Sustainability*, 14(21), 14507.
<https://doi.org/10.3390/su142114507>
- Xiaomei, Z., Rashdan Sallehuddin, M., Mohd Saad, R., & Lu, Z. (2021). Gender diversity, political connections and environmental, social and governance performance in china: the moderating role of tenure. *Studies of Applied Economics*, 39(10).
<https://doi.org/10.25115/eea.v39i10.5962>
- Xie, J., Nozawa, W., Yagi, M., Fujii, H., & Managi, S. (2019). Do environmental, social, and governance activities improve corporate financial performance? *Business Strategy and the Environment*, 28(2), 286–300. <https://doi.org/10.1002/bse.2224>
- Yarram, S. R., & Adapa, S. (2021). Board gender diversity and corporate social responsibility: Is there a case for critical mass? *Journal of Cleaner Production*, 278, 123319.
<https://doi.org/10.1016/j.jclepro.2020.123319>
- Zadek, S. (2007). The path to corporate responsibility. In W. C. Zimmerli, M. Holzinger, & K. Richter (Eds.), *Corporate Ethics and Corporate Governance* (pp. 159–172). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-540-70818-6_13
- Zhang, W., Luo, Q., & Liu, S. (2022). Is government regulation a push for corporate environmental performance? Evidence from China. *Economic Analysis and Policy*, 74, 105–121. <https://doi.org/10.1016/j.eap.2022.01.018>

CHAPTER VII

APPENDICIES

Appendix A

Ballotpedia-Legiscan ESG Legislation

This appendix provides a comprehensive catalog of state-level ESG legislation enacted between 2020 and 2024 in the United States. The data were systematically compiled from two authoritative sources: LegiScan and Ballotpedia's ESG Tracker. Ballotpedia's ESG Tracker served as the primary identification mechanism for relevant legislation, while LegiScan provided access to the complete legislative texts for detailed analysis.

Table A1. Ballotpedia/Legiscan ESG Legislation

STATE	BILL	YEAR	TYPE	KEY PROVISIONS
ALABAMA	SB261	2023	Anti	Prohibits state entities from contracting with companies that boycott fossil fuel, timber, mining, agriculture, or firearms industries
ALABAMA	SB129	2024	Anti	Prohibits public institutions from maintaining DEI offices and promoting "divisive concepts"
ARKANSAS	HB1253	2023	Anti	Prohibits consideration of non-pecuniary factors in state pension fund management decisions
ARKANSAS	HB1845	2023	Anti	Creates ESG Oversight Committee to identify financial services providers that "discriminate" against energy and firearms industries
ARKANSAS	SB62	2023	Anti	Prohibits public entities from contracting with companies that boycott energy, fossil fuel, firearms, and ammunition industries
ARKANSAS	HB1307	2023	Anti	Authorizes state treasurer to divest from financial institutions that discriminate against energy, fossil fuel, and firearms industries
ARKANSAS	HB1637	2021	Anti	Establishes mechanism for state to resist federal executive orders related to environmental regulation and ESG standards

FLORIDA	H0003	2023	Anti	Requires state financial decisions to be based solely on financial factors rather than ESG considerations
FLORIDA	H0005	2023	Anti	Expands restrictions on state investments related to companies with business operations in Iran
FLORIDA	H0989	2024	Anti	Prohibits financial institutions from denying services based on political opinions, religious beliefs, or business sector
GEORGIA	HB481	2024	Anti	Establishes strict sole fiduciary framework prohibiting consideration of "nonpecuniary interests" in pension management
GEORGIA	HB1018	2024	Anti	Prohibits financial institutions from using specific merchant codes to identify firearms retailers
IDAHO	S1405	2022	Anti	Prohibits consideration of nonpecuniary factors in state investments unless they directly impact financial returns
IDAHO	H0190	2023	Anti	Requires financial institutions to certify they don't boycott fossil fuel, timber, minerals, agriculture, or firearms industries
IDAHO	S1291	2024	Anti	Prohibits public entities from contracting with companies that boycott specific industries including fossil fuels
IDAHO	H0191	2023	Anti	Prohibits consideration of ESG criteria in state procurement decisions
IDAHO	S1274	2024	Anti	Prohibits public postsecondary institutions from requiring diversity statements in hiring processes
IDAHO	H0691	2024	Anti	Requires public disclosure for significant state financial agreements over \$25 million
INDIANA	HB1008	2023	Anti	Prohibits state pension system from making investment decisions for non-financial purposes

IOWA	SF2435	2024	Anti	Prohibits public higher education institutions from establishing DEI offices or requiring diversity statements
LOUISIANA	HCR110	2023	Anti	Urges retirement systems to avoid ESG considerations in investment decisions
LOUISIANA	HR267	2024	Anti	Requires retirement systems to submit reports on proxy voting activities
LOUISIANA	HCR70	2023	Anti	Requires reporting on companies that "discriminate against" fossil fuel industry through ESG policies
LOUISIANA	SB234	2024	Anti	Prohibits state entities from contracting with companies that discriminate against firearm industries
LOUISIANA	HR203	2022	Anti	Establishes task force to study regulation of ESG criteria in lending and investment practices
LOUISIANA	HR246	2022	Anti	Creates study group to examine ESG factors in lending and investment practices
LOUISIANA	HCR78	2024	Anti	Requires public higher education institutions to justify their use of ESG criteria
LOUISIANA	HCR59	2023	Anti	Formally opposes SEC's proposed climate disclosure requirements
MISSISSIPPI	SB2226	2024	Anti	Extends prohibition against state investment in companies that boycott Israel
MISSOURI	HR12	2023	Anti	Opposes federal climate-related financial disclosure requirements and ESG mandates
MONTANA	HB228	2023	Anti	Prohibits consideration of ESG factors in state investment decisions unless they directly impact financial returns

MONTANA	HB356	2023	Anti	Prohibits state entities from contracting with companies that discriminate against firearms businesses
MONTANA	HJ11	2023	Anti	Urges Congress to oppose ESG policies deemed harmful to resource-based economy
NEBRASKA	LB686	2024	Anti	Establishes governance framework for firefighter retirement systems with fiduciary responsibility focus
NORTH DAKOTA	SB2291	2021	Anti	Prohibits state investment board from making "social investments" without financial justification
NORTH DAKOTA	HB1429	2023	Anti	Prohibits state entities from making "social investments" and insurance companies from discriminating based on ESG criteria
NEW HAMPSHIRE	HB457	2023	Anti	Requires state treasurer to prioritize financial benefits and report on funds with "mixed motivations"
NEW HAMPSHIRE	HB1469	2022	Anti	Prohibits banks from using social credit scores or ESG criteria in business decisions
OKLAHOMA	HB2034	2022	Anti	Prohibits state investment in financial companies that "boycott" energy companies
OKLAHOMA	HB1236	2021	Anti	Establishes mechanism to review and potentially reject federal actions related to ESG standards
SOUTH CAROLINA	H3690	2024	Anti	Prohibits consideration of environmental, social, or political factors in pension investment decisions
TENNESSEE	SB0955	2023	Anti	Establishes state treasurer as sole fiduciary authority over state investment decisions
TENNESSEE	HB1286	2023	Anti	Requires state treasurer to make investment decisions based solely on financial factors

TENNESSEE	HB2672	2022	Anti	Prohibits state treasurer from contracting with financial institutions that have policies against financing fossil fuel companies
TENNESSEE	SB2649	2022	Anti	Prohibits state treasurer from contracting with financial institutions that have policies against financing fossil fuel companies
TENNESSEE	SB2148	2024	Anti	Prevents financial institutions from discriminating against customers based on non-financial factors
TENNESSEE	HB2100	2024	Anti	Prohibits financial institutions from denying services based on political opinions or business sector
TENNESSEE	SB2223	2024	Anti	Prohibits financial institutions from tracking firearm and ammunition purchases through merchant codes
TEXAS	SB19	2021	Anti	Prohibits state contracts with companies that "discriminate" against firearms businesses
TEXAS	SB833	2023	Anti	Prohibits insurers from using ESG criteria for pricing without sound actuarial principles
UTAH	SB0096	2023	Anti	Requires public entities to make investment decisions solely to maximize risk-adjusted returns
UTAH	SCR009	2023	Anti	Opposes ESG investing practices within state financial operations
UTAH	SB0097	2023	Anti	Prohibits state entities from contracting with companies engaged in "economic boycotts" of specific industries
UTAH	HB0449	2023	Anti	Prohibits coordinated boycotts against companies in specified sectors including fossil fuels
UTAH	HB0281	2023	Anti	Prohibits governmental entities from using social credit scores

UTAH	HB0415	2021	Anti	Restricts implementation of federal executive orders related to ESG standards
WEST VIRGINIA	HB2862	2023	Anti	Prohibits consideration of ESG factors in investment decisions unless they directly affect financial returns
WEST VIRGINIA	SB262	2022	Anti	Authorizes creation of a list of financial institutions boycotting energy companies
WYOMING	HB0236	2021	Anti	Prohibits financial institutions from discriminating against firearms businesses
CALIFORNIA	SB1089	2022	Pro	Extends prohibition on investing in Turkish government securities due to human rights concerns
CALIFORNIA	AB979	2020	Pro	Requires public corporations to include directors from underrepresented communities on boards
CALIFORNIA	SB261	2023	Pro	Requires corporations with revenue over \$500 million to disclose climate-related financial risks
CALIFORNIA	SB253	2023	Pro	Requires large corporations to publicly disclose greenhouse gas emissions
CALIFORNIA	SB655	2021	Pro	Strengthens diversity requirements for insurers operating in California
COLORADO	SB016	2023	Pro	Strengthens greenhouse gas reduction targets and requires climate risk disclosure
DELAWARE	HB334	2020	Pro	Authorizes consideration of social, environmental, and governance values in institutional fund management
ILLINOIS	SB2152	2023	Pro	Modifies pension code regarding proxy voting and sustainability reporting

ILLINOIS	HB2782	2023	Pro	Mandates integration of sustainability factors into public investment decisions
ILLINOIS	HB1293	2022	Pro	Requires pension funds to divest from Russian and Belarusian investments
ILLINOIS	SB1730	2021	Pro	Expands corporate board diversity reporting to include LGBTQ individuals
ILLINOIS	SB1699	2023	Pro	Establishes parameters for renewable energy facilities and thermal energy networks
ILLINOIS	SB0653	2021	Pro	Expands State Treasurer's authority to consider ESG factors in financial decisions
MAINE	LD99	2021	Pro	Requires divestment of state pension funds from fossil fuel companies by 2026
MAINE	LD1895	2023	Pro	Establishes framework for offshore wind energy development with environmental and social considerations
MARYLAND	SB566	2022	Pro	Requires pension fiduciaries to consider climate-related risks in investment decisions
MARYLAND	HB740	2022	Pro	Requires pension fiduciaries to consider climate-related risks in investment decisions
MARYLAND	HB1212	2024	Pro	Establishes Director of DEI and governance specialist positions in state retirement agency
MINNESOTA	HF4165	2022	Pro	Requires divestment from Russian and Belarusian assets in response to Ukraine invasion
OREGON	HB4083	2024	Pro	Requires divestment from thermal coal companies in state investment portfolio

OREGON	HB2021	2021	Pro	Establishes greenhouse gas reduction targets for electricity sector
OREGON	SB1047	2023	Pro	Promotes diversity and inclusion in public procurement processes
RHODE ISLAND	H7127	2024	Pro	Establishes state-administered retirement savings program with ESG investment options
RHODE ISLAND	S2045	2024	Pro	Establishes state-administered retirement savings program with ESG investment options
RHODE ISLAND	S2583	2022	Pro	Mandates procurement of offshore wind capacity aligned with state climate goals
RHODE ISLAND	H7971	2022	Pro	Mandates procurement of offshore wind capacity with environmental and labor protections

Note: This table catalogs state-level ESG legislation in the United States from 2020-2024. "Pro-ESG" bills mandate or encourage consideration of environmental, social, or governance factors in investment decisions, corporate operations, or public policy. "Anti-ESG" bills explicitly restrict or prohibit such considerations, typically requiring decisions based solely on financial factors. Because this study used Ballotpedia's ESG tracker, legislation that might implicate fringe ESG topics are included.

Appendix B

Variable Calculations and Definitions

This appendix provides detailed information on the construction of key financial variables used in the analysis. All variables are calculated using data from Compustat annual database for the years 2019-2024

Table B1. Financial Control Variable Calculations and Compustat Codes

Measure	Sub Variables
$\text{Leverage} = \frac{DLTT + DLC}{SEQ}$	$DLTT$ = Long Term Debt DLC = Debt in Current Liabilities SEQ = Indicates Shareholders' Equity
$\text{Firm Size} = \log(AT)$	AT = Total Assets
$ROA = \frac{NI}{AT}$	NI = Net Income AT = Total Assets
$\text{Tobin's } Q = \frac{AT + (CSHO \times PRCC_F) - CEQ}{AT}$	$CSHO$ = Common Shares Outstanding $PRCC_F$ = Fiscal Year Closing Share Price CEQ = Common Equity AT = Total Assets

Note: Firms with negative leverage and ROA values were excluded from the analysis as these typically represent distressed companies with negative equity, potentially distorting financial relationship interpretations.